

# PDR WORLD

## MASTER PRODUCT CATALOGUE 2024-2025

### Empowering Global Connectivity

Discover our comprehensive suite of state-of-the-art optical fiber solutions, carrier-grade passive and active components, advanced cable management systems, and precision testing equipment.

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ISO 9001:2015 & 14001:2015 Certified

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## ACTIVE COMPONENTS &gt; ACTIVE OPTICAL CABLE (AOC)

# Active Optical Cable (AOC)

*100G QSFP28 active optical cable, a reliable, low-power alternative to transceiver assemblies.*

PDR 100G QSFP28 Active Optical Cables are Ethernet, InfiniBand and MSA compliant, feature DDM, secure pull-tab latching, slim 3.0 mm OM3 cable, and are available in lengths from 0.5 m to 100 m.

## Key Features

- Supports 100G across 4 bi-directional lanes at 25.78 Gbps per channel
- Low power consumption, maximum 1.9 W per end
- Digital Diagnostic Monitoring (DDM/DOM) with I<sup>2</sup>C real-time alerts
- Available in lengths from 0.5 m to 100 m
- Hot-swappable QSFP28 form factor with high-strength pull-tab latch
- Flexible OM3 multimode fiber with slim 3.0 mm outer diameter

## Applications

- Point-to-point connections within data centres
- Rack-to-rack and inter-row top-of-rack architectures
- High-performance computing and storage racks

## Technical Specifications

| Parameter / Specification | Value / Detail                                              |
|---------------------------|-------------------------------------------------------------|
| Form Factor               | QSFP28 (hot-swappable)                                      |
| Data Rate                 | 100 Gbps (4 lanes × 25.78 Gbps)                             |
| Operating Wavelength      | 850 nm                                                      |
| Fiber Type                | OM3 multimode                                               |
| Cable Diameter (OD)       | 3.0 mm ± 0.20                                               |
| Available Lengths         | 0.5 m to 100 m                                              |
| Max Power Consumption     | 1.9 W per end                                               |
| Operating Temperature     | 0°C to 70°C                                                 |
| Compliance                | QSFP28 MSA, IEEE P802.3bm, 100GBASE-SR4, RoHS/REACH, TÜV/UL |

## PASSIVE COMPONENTS

# Variable Fiber Attenuator

*Variable optical attenuator for precise in-line signal level control.*

PDR Variable Fiber Optic Attenuator (VOA) provides continuously adjustable attenuation up to 30 dB for single-mode and multimode fiber with accurate, low-loss performance across 1260–1650 nm for telecom, CATV, data center, and test applications.

## Key Features

- Wide attenuation range of 0 to 30 dB with precise, continuous adjustment capability
- Low insertion loss of  $\leq 1.0$  dB to minimise signal degradation in the optical path
- Available in SC, LC, FC, and ST connector interfaces for universal compatibility
- High attenuation accuracy of  $\pm 0.5$  dB for reliable signal level management
- High return loss of  $\geq 50$  dB (SM) for excellent back-reflection suppression
- Supports both single-mode and multi-mode fiber types

## Applications

- Fiber-to-the-Home (FTTH) and Fiber-to-the-Building (FTTB) network power balancing
- Cable TV (CATV) and broadband infrastructure signal level optimisation
- Telecommunications network testing, commissioning, and maintenance
- Data centre interconnects and optical link budget management

## Technical Specifications

| Parameter / Specification         | Value / Detail           |
|-----------------------------------|--------------------------|
| Attenuation Range                 | 0 to 30 dB (Typical)     |
| Attenuation Accuracy              | ± 0.5 dB                 |
| Resolution                        | 0.1 dB (Typical)         |
| Insertion Loss                    | ≤ 1.0 dB                 |
| Return Loss                       | ≥ 50 dB (SM)             |
| Wavelength Range                  | 1260 to 1650 nm          |
| Connector Type                    | SC / LC / FC / ST        |
| Fiber Type                        | Single-mode / Multi-mode |
| Polarisation Dependent Loss (PDL) | ≤ 0.2 dB                 |
| Repeatability                     | ≤ ± 0.2 dB               |
| Operating Temperature             | -20°C to +70°C           |
| Storage Temperature               | 40°C to +85°C            |
| Durability                        | ≥ 1000 adjustment cycles |

## PASSIVE COMPONENTS

# Bare Fiber Adapter

*Bare fiber adapter for quickly coupling unterminated fiber to test equipment.*

PDR Bare Fiber Adapters enable tool-free temporary connection of stripped bare fibers to SC, LC, FC, ST, or SMA interfaces for reliable testing, field, and lab applications.

## Key Features

- Available in SC, LC, FC, ST, and SMA interface types to suit all common test equipment
- Compatible with both single-mode (OS1/OS2) and multimode (OM1, OM4) optical fibers
- Simple plug-and-play operation, insert bare fiber on one side, connect to equipment on the other
- Secure mechanical clamping mechanism holds bare fiber firmly without adhesive or tools
- Compact, lightweight body for easy handling in lab and field environments
- Enables fast temporary termination for fiber testing and measurement applications

## Applications

- Fiber-to-the-Home (FTTH) and Fiber-to-the-Building (FTTB) network testing and commissioning
- OTDR and optical power meter connections during cable installation and characterisation
- Telecommunications network measurement, fault location, and maintenance
- Data centre and enterprise LAN fiber testing and troubleshooting

## Technical Specifications

| Parameter / Specification | Value / Detail                                                      |
|---------------------------|---------------------------------------------------------------------|
| Adapter Types             | SC, LC, FC, ST, SMA                                                 |
| Compatible Fiber          | Single-mode (OS1/OS2)   Multimode (OM1, OM2, OM3, OM4)              |
| Fiber Cladding Diameter   | 125 μm (standard)                                                   |
| Fiber Holding Mechanism   | Mechanical clamp, no adhesive or tools required                     |
| Insertion Loss            | SC/FC/ST: ≤ 0.5 dB   LC: ≤ 0.5 dB   SMA: ≤ 1.0 dB (typical)         |
| Return Loss               | SC/FC/ST: ≥ 20 dB   LC: ≥ 20 dB   SMA: ≥ 14 dB                      |
| Mating Durability         | ≥ 500 mating cycles                                                 |
| Housing Material          | Zinc alloy body with precision ceramic / stainless insert           |
| Operating Temperature     | 20°C to +70°C                                                       |
| Storage Temperature       | -40°C to +85°C                                                      |
| Operating Humidity        | 5%, 95% RH (non-condensing)                                         |
| Bare Fiber Insertion      | Simple straight insertion, no stripping jig required                |
| Applications              | Lab testing, field measurement, emergency restoration, PON/OTDR use |
| Compliance                | RoHS compliant                                                      |

## PASSIVE COMPONENTS

# Cassette Cleaner

*Reel cassette cleaner for safe, lint-free connector end-face cleaning.*

PDR Fiber Optic Connector Cleaner: Compact handheld cleaner for SC, FC, ST, MU, LC, MT, D4, DIN & E2000 connectors, using non-alcohol superfine fiber tape for scratch-free, residue-free cleaning with 500+ cycles per replaceable cartridge.

## Key Features

- Compatible with SC, FC, ST, MU, LC, MT, D4, DIN, and E2000 fiber optic connectors
- Anti-static design prevents re-contamination after each cleaning stroke
- Replaceable tape cartridge design minimises ongoing consumable costs
- Special non-alcohol superfine fiber tape for safe, residue-free end-face cleaning
- Over 500 cleaning cycles per tape cartridge, low cost per clean
- Simple one-push action for consistent, repeatable cleaning results

## Applications

- Fiber-to-the-Home (FTTH) and Fiber-to-the-Building (FTTB) connector maintenance
- Data centre patch panel, switch port, and transceiver connector cleaning
- Telecommunications equipment room and outside plant installation and servicing
- Cable TV (CATV) and broadband network end-face preparation and maintenance

## Technical Specifications

| Parameter / Specification    | Value / Detail                         |
|------------------------------|----------------------------------------|
| Compatible Connectors        | SC, FC, ST, MU, LC, MT, D4, DIN, E2000 |
| Cleaning Medium              | Non-alcohol superfine fiber tape       |
| Cleaning Times per Cartridge | 500+ cycles                            |
| Replacement Cartridge        | Available (field-replaceable)          |
| Anti-static                  | Yes                                    |
| Cleaning Action              | One-push mechanical advance            |
| Dimensions (L×W×H)           | 115 mm × 79 mm × 32 mm                 |
| Weight                       | Lightweight, suitable for field carry  |
| Solvents / Liquids Required  | None                                   |

## CABLE MANAGEMENT DEVICES

# CAT 6 Patch Panel

*24-port Cat 6 patch panel for organised copper network termination.*

PDR Cat 6 24-Port Distribution Panel features 24 RJ-45 IDC ports with T568A/T568B support, meeting ISO/IEC 11801 and EIA/TIA standards for Gigabit and 10GBase-T structured cabling.

## Key Features

- 24-port RJ-45 panel conforming to ISO/IEC 11801 and EIA/TIA 568A & 568B standards
- Supports Cat 6 performance, suitable for Gigabit Ethernet and 10G-Base-T applications
- Metallic panel (SPCC) with fireproof ABS UL-94V0 plastic body for durability and safety
- RJ-45 jack housing: PBT / 8P8C with 50 micro-inch gold-plated contacts for reliable connectivity
- PCB: FR-4 / 1.6 mm UL-94V0 for superior signal integrity
- Supports both T568A and T568B colour-coding with clearly labelled port identification

## Applications

- Data centre patch rooms and server rack cable management
- Enterprise LAN and structured cabling infrastructure
- Office building and campus network distribution frames
- Telecommunications rooms and main distribution frames (MDF / IDF)

Technical Specifications

| Parameter / Specification | Value / Detail                                 |
|---------------------------|------------------------------------------------|
| Standard Compliance       | ISO/IEC 11801   EIA/TIA 568A & 568B            |
| Number of Ports           | 24 x RJ-45                                     |
| Category                  | Cat 6                                          |
| Panel Material            | Metallic, SPCC (Steel, Cold-rolled, Coil)      |
| Plastic Body              | Fireproof ABS UL-94V0                          |
| RJ-45 Jack Housing        | PBT / 8P8C, 50 micro-inch gold plated contacts |
| PCB                       | FR-4 / 1.6 mm UL-94V0                          |
| IDC Body                  | Fireproof ABS UL-94V0                          |
| IDC Pin Material          | Phosphor Bronze, Tinned                        |
| IDC Wire Range            | 22 to 26 AWG solid conductor                   |
| Termination Tool          | 110 or Krone punch-down tool                   |
| Color Coding              | T568A and T568B (Arrange: 8 7 6 3 2 1 4 5)     |
| Insulation Resistance     | > 500 MΩ (minimum)                             |
| DC Resistance             | < 0.1 Ω (maximum)                              |

## PASSIVE COMPONENTS

# CAT 6 Patch Cord

*High-speed copper patch cord*

The PDR U/UTP Cat 6 Patch Cord provides flexible, reliable high-speed Ethernet connectivity with Cat 6 RJ45 connectors and PVC or LSZH jacket options.

## Key Features

- U/UTP 4-pair construction with stranded 24 AWG (7×0.18 mm) conductors for flexibility and durability
- Cat 6 RJ45 plug termination with moulded PVC strain relief boot, colour-matched to outer jacket
- Jacket outer diameter 5.8 ± 0.20 mm with ripcord for easy stripping during termination
- HDPE insulation for low capacitance and consistent electrical performance across all four pairs
- Outer jacket available in PVC or LSZH, specify at time of order to suit installation requirements
- Pair colour coding: Blue/White, Orange/White, Green/White, Brown/White

## Applications

- 100 Base-TX and 100 Base-T (100VG-AnyLAN) Fast Ethernet network connections
- 1000 Base-TX high-speed enterprise LAN patch panel deployments
- 1000 Base-T Gigabit Ethernet structured cabling installations
- Data centre patch panel and server rack interconnections requiring flexible, durable patch cords

## Technical Specifications

| Parameter / Specification | Value / Detail                                     |
|---------------------------|----------------------------------------------------|
| Construction              | U/UTP (Unshielded Twisted Pair)                    |
| Pair Count                | 4 pairs                                            |
| Conductor                 | Stranded 24 AWG (7×0.18 mm)                        |
| Insulation Material       | HDPE                                               |
| Insulation Diameter       | 0.91 ± 0.01 mm                                     |
| Pair Colours              | Blue/White, Orange/White, Green/White, Brown/White |
| Cross Filler              | Yes                                                |
| Connector                 | Cat 6 RJ45 Plug                                    |
| Strain Relief Boot        | Moulded PVC, colour-matched to jacket              |
| Outer Jacket Material     | PVC or LSZH (specify at order)                     |
| Jacket Thickness          | 0.6 ± 0.05 mm                                      |
| Outer Diameter            | 5.8 ± 0.20 mm                                      |
| Ripcord                   | Yes                                                |
| Rated Temperature         | 60°C, 75°C, 90°C                                   |

## PASSIVE COMPONENTS

# Fiber Optic Cleaner Pen

*One-click cleaner pen for fast, reliable connector end-face cleaning.*

PDR Fiber Optic Cleaner Pen is a compact, single-action cleaner for female connectors, removing dust, oil, and contaminants without scratching, with anti-static microfiber technology for 800+ cleanings and optimal optical performance.

## Key Features

- Push-to-clean operation, easy single-motion engagement initiates the cleaning cycle
- Anti-static resin body, prevents electrostatic charge build-up during cleaning
- Extendable tip design reaches recessed connectors on panels and equipment
- 800+ cleanings per unit, cost-effective and long lasting
- Densely stranded, debris-free micro-fiber cleaning element for thorough end-face cleaning
- Cleaning system rotates 180° for a full, complete sweep of the ferrule end face

## Applications

- Fiber network panels and patch panel assemblies in data centres
- Cable assembly production facilities requiring consistent end-face cleanliness
- Outdoor FTTX (Fiber-to-the-X) network installations and field maintenance
- Testing laboratories and quality-control environments for fiber optic components

## Technical Specifications

| Parameter / Specification | Value / Detail                                              |
|---------------------------|-------------------------------------------------------------|
| Compatible Connectors     | 2.5 mm: FC, SC, ST   1.25 mm: LC, MU   2.0 mm: SMPTE        |
| Ferrule Tip Diameter      | 2.5 mm (CLEP 2.5)   1.25 mm (CLEP 1.25)   2.0 mm (CLEP 2.0) |
| Cleaning Method           | Push-to-clean rotary mechanism                              |
| Cleaning Rotation         | 180° full sweep per actuation                               |
| Cleanings Per Unit        | 800+ cleanings                                              |
| Cleaning Element          | Densely stranded micro-fiber, debris-free                   |
| Tip Design                | Extendable, reaches recessed connectors                     |
| Engagement Indicator      | Audible click on correct engagement                         |
| Body Material             | Anti-static resin                                           |
| Operation                 | Single push motion, no batteries required                   |
| Disposal                  | Disposable, replace when cleanings are exhausted            |

## PASSIVE COMPONENTS

# CPRI Patchcord

*Outdoor waterproof fiber optic assembly*

PDR CPRI Outdoor Waterproof Patch Cords feature rugged IP-rated, blind-mate fiber assemblies with multiple waterproof connector options, armored/non-armored cable designs, and 2–24 core single-mode or multimode fiber for harsh outdoor applications.

## Key Features

- Waterproof connector housings: ODVA, IP, FX, PDLC, and NSN types with MPO/MTP, LC, or SC ferrules
- Robust bayonet locking mechanism for easy, fast, and secure connector mating and retention
- Available in flat drop or round cable, armored or non-armored, for flexible routing requirements
- IP-rated blind-mating design for quick, tool-free installation in field and outdoor environments
- Lightweight, waterproof, dustproof, and corrosion-resistant construction for harsh environments
- Single mode fiber: G.652D, G.657A1, G.657A2, G.657B3, multimode: OM1, OM2, OM3, OM4

## Applications

- WiMAX and LTE base station fiber backhaul and fronthaul cabling in outdoor environments
- Industrial outdoor installations in factories, plants, and process control environments
- Remote Radio Head (RRH) and Remote Radio Unit (RRU) interconnections in distributed antenna systems
- Harsh environment deployments requiring waterproof, dustproof, and corrosion-resistant terminations

Technical Specifications

| Parameter / Specification  | Value / Detail                             |
|----------------------------|--------------------------------------------|
| Cable Diameter             | 5.0 mm   7.0 mm   8.0 mm (±0.3 mm)         |
| Connector Housing Types    | ODVA   IP   FX   PDLC   NSN                |
| Ferrule / Connector Types  | MPO/MTP   LC   SC                          |
| Fiber Mode, Single-Mode    | G.652D   G.657A1   G.657A2   G.657B3       |
| Fiber Mode, Multimode      | OM1   OM2   OM3   OM4                      |
| Fiber Count                | 2 / 4 / 8 / 12 / 24 core                   |
| Cable Jacket               | LSZH   TPU   PE                            |
| Cable Length               | 1 to 150 m; custom lengths available       |
| Operating Temperature      | -40°C to +85°C                             |
| SM Insertion Loss, Typical | ≤0.30 dB (IEC 61300-3-34)                  |
| SM Insertion Loss, Maximum | ≤0.75 dB (IEC 61300-3-34)                  |
| MM Insertion Loss, Typical | ≤0.50 dB (IEC 61300-3-34)                  |
| MM Insertion Loss, Maximum | ≤0.75 dB (IEC 61300-3-34)                  |
| SM Return Loss             | ≥50 dB (PC)   ≥60 dB (APC) (IEC 61300-3-6) |

## PASSIVE COMPONENTS

# CWDM Mux / DeMux Module

*CWDM multiplexer / demultiplexer modules for cost-effective wavelength expansion.*

PDR CWDM Mux/DeMux offers 4/8/16-channel CWDM (1270–1610 nm) in ABS, LGX, and 1U rack formats with low loss, high isolation, and reliable optical performance.

## Key Features

- Four form factors: ABS Compact Module (two sizes), 1U Rack Equipment, and LGX Module
- Low insertion loss:  $\leq 1.6$  dB (4ch),  $\leq 2.8$  dB (8ch),  $\leq 4.5$  dB (16ch) for ABS Compact Module
- Low passband ripple  $\leq 0.40$  dB and polarization dependent loss  $< 0.30$  dB across all variants
- Supports 4, 8, and 16 channel configurations across the CWDM grid from 1270 nm to 1610 nm
- High adjacent channel isolation  $> 30$  dB; non-adjacent channel isolation  $> 40$  dB
- Return loss  $> 45$  dB and directivity  $> 50$  dB for excellent signal integrity

## Applications

- CATV and cable television headend signal multiplexing and distribution across CWDM wavelength bands
- Optical network monitoring and per-wavelength channel management in carrier-grade systems
- Metro and access WDM network capacity expansion using the full 1270 to 1610 nm CWDM grid
- Test and measurement equipment for wavelength-selective optical characterisation

# Technical Specifications

| Parameter / Specification          | Value / Detail                                                   |
|------------------------------------|------------------------------------------------------------------|
| CWDM Wavelength Grid               | 1270, 1290, 1310 ... 1570, 1590, 1610 nm (20 nm channel spacing) |
| Channel Configurations             | 4 Channel   8 Channel   16 Channel                               |
| IL, ABS Compact Module (Narrow PB) | 4ch: ≤1.6 dB   8ch: ≤2.8 dB   16ch: ≤4.5 dB                      |
| IL, ABS Compact Module (Wide PB)   | 4ch: ≤2.0 dB   8ch: ≤3.6 dB   16ch: ≤5.0 dB                      |
| IL, 1U Rack Equipment              | 4ch: ≤1.8 dB   8ch: ≤3.0 dB   16ch: ≤4.8 dB                      |
| IL, LGX Module                     | 4ch: ≤1.8 dB   8ch: ≤3.0 dB   16ch: ≤4.8 dB                      |
| Passband, ABS Compact (Narrow PB)  | ±0.11 nm (@-0.5 dB)                                              |
| Passband, ABS Compact (Wide PB)    | ±6.5 nm (@-0.5 dB)                                               |
| Passband, 1U Rack / LGX Module     | ±7.5 nm (@-0.5 dB)                                               |
| Isolation, Adjacent Channel        | > 30 dB                                                          |
| Isolation, Non-Adjacent Channel    | > 40 dB                                                          |
| Passband Ripple                    | ≤0.40 dB                                                         |
| Polarization Dependent Loss (PDL)  | < 0.30 dB                                                        |

| Parameter / Specification          | Value / Detail |
|------------------------------------|----------------|
| Polarization Mode Dispersion (PMD) | < 0.10 ps      |

## ACTIVE COMPONENTS &gt; DIRECT ATTACHED CABLE (DAC)

# Direct Attached Cable (DAC)

*Passive SFP+ twinax copper cable for short-reach 10G interconnects, zero power, zero heat.*

PDR SFP+ Direct Attach Cables are passive, hot-swappable, MSA-compliant 10 Gbps twinax copper cables for reliable short-reach connectivity, factory-tested and RoHS/REACH compliant.

## Key Features

- Passive twinax copper, zero power, no heat generation
- Supports 10 Gbps (10GBASE-CR, 10G FCoE, 10G InfiniBand)
- Broad compatibility with switches, servers and storage
- Available in 1 m, 2 m, 3 m and 5 m lengths
- Hot-swappable SFP+ connectors, SFP+ MSA compliant
- 100% factory-tested before shipment

## Applications

- Top-of-rack and rack-to-rack switching
- Data-centre and HPC short-reach interconnects
- Enterprise server-to-switch links

## Technical Specifications

| Parameter / Specification | Value / Detail                                    |
|---------------------------|---------------------------------------------------|
| Cable Type                | Passive twinax copper                             |
| Connector Type            | SFP+ (both ends)                                  |
| Nominal Data Rate         | 10.3125 Gbps                                      |
| Available Lengths         | 1 m   2 m   3 m   5 m                             |
| Link Distance             | Up to 7 m (passive)                               |
| Minimum Bend Radius       | 30 mm                                             |
| Operating Temperature     | 0°C to 70°C                                       |
| Power Consumption         | Passive, zero power draw                          |
| Compliance                | SFP+ MSA, IEEE 802.3ae, 10GFC, RoHS/REACH, TÜV/UL |

## SPECIALTY DRONES

## Ground unit

*Ground-side fiber optical terminal that outputs video and links to the remote controller, with selectable CRSF/SBUS protocols.*

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Ground-side receive terminal that recovers the 1.25 Gbps single-fiber link and outputs composite video to a display while interfacing with the remote controller, image-transmission module or FPV receiver, with dial-code selectable GH1.25 or cable modes and CRSF/SBUS protocol support.

### Key Features

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- Ground-side (Type B) receive terminal for tethered / FPV drone fiber links
- Composite video output via BNC to display or FPV receiver
- Selectable CRSF or SBUS protocol for popular FPV / racing-drone flight controllers
- Compact 105 × 70 × 28 mm housing
- Single-fiber optical reception at 1.25 Gbps over an FC interface
- Dial-code selectable GH1.25 (W) or cable (C) interface mode
- FC optical, RJ45 100M Ethernet and BNC video front-panel interfaces

### Applications

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- Defence ISR & secure communications
- FPV reconnaissance & tactical missions
- Perimeter / border surveillance
- EW / jam-resistant operations

# Technical Specifications

| Parameter / Specification | Value / Detail                                |
|---------------------------|-----------------------------------------------|
| Unit Type                 | Type B — Ground Unit                          |
| Optical Transmission Rate | 1.25 Gbps                                     |
| Optical Connector         | FC (single fiber)                             |
| Video Channel             | Composite video output (BNC)                  |
| Serial / Telemetry        | TTL serial, PPM input; CRSF / SBUS support    |
| Interface Modes           | GH1.25 (W) or Cable (C), dial-code selectable |
| Power Input               | 10–26 V DC (DC-IN); GH1.25 outputs 5 V        |

## PASSIVE COMPONENTS

# DWDM Mux/DeMux Module

*DWDM multiplexer / demultiplexer modules for high-capacity dense wavelength transport.*

PDR DWDM Mux/DeMux supports 4/8/16-channel ITU-T C20–C60 DWDM (1520–1570 nm, 100 GHz) in ABS or 1U rack formats with low loss, >30 dB isolation, and >45 dB return loss.

## Key Features

- Available in compact ABS module (100×80×10 mm / 120×80×18 mm) and 1U rack equipment form factors
- Supports 4, 8, and 16 channel configurations covering ITU-T channels C20 to C60
- Low insertion loss: ≤2.0 dB (4ch), ≤3.0 dB (8ch), ≤4.8 dB (16ch) for ABS module
- Operating wavelength range 1520 to 1570 nm with 100 GHz minimum channel spacing
- High channel isolation: >30 dB adjacent channel, >40 dB non-adjacent channel
- Low passband ripple (≤0.30 dB) and polarization dependent loss (≤0.30 dB)

## Applications

- CATV and cable television headend signal multiplexing and distribution
- Dense WDM (DWDM) transport systems for long-haul and metro network capacity expansion
- Optical network monitoring and wavelength management in carrier-grade systems
- Test and measurement equipment for wavelength-selective optical characterisation

# Technical Specifications

| Parameter / Specification          | Value / Detail                              |
|------------------------------------|---------------------------------------------|
| Channel Configurations             | 4 Channel   8 Channel   16 Channel          |
| Insertion Loss, ABS Module         | 4ch: ≤2.0 dB   8ch: ≤3.0 dB   16ch: ≤4.8 dB |
| Insertion Loss, 1U Rack Equipment  | 4ch: ≤2.2 dB   8ch: ≤3.2 dB   16ch: ≤5.0 dB |
| ITU Channels                       | C20, C60 (C20, C21, C22 ... C58, C59, C60)  |
| Operating Wavelength               | 1520 to 1570 nm                             |
| Min. Channel Spacing               | 100 GHz                                     |
| Passband (−0.5 dB bandwidth)       | ±0.11 nm                                    |
| Isolation, Adjacent Channel        | > 30 dB                                     |
| Isolation, Non-Adjacent Channel    | > 40 dB                                     |
| Passband Ripple                    | ≤0.30 dB                                    |
| Polarization Dependent Loss (PDL)  | ≤0.30 dB                                    |
| Polarization Mode Dispersion (PMD) | < 0.10 ps                                   |
| Return Loss                        | > 45 dB                                     |
| Directivity                        | > 50 dB                                     |

TEST AND MEASUREMENT EQUIPMENT

# EasyGet WiFi Wireless Fiber Endface Microscope

*Wireless transmission to smartphone/tablet*

Wireless transmission to smartphone/tablet

## Key Features

- Wireless transmission to smartphone/tablet
- Eliminates direct eye exposure to lasers
- High-resolution endface imaging

## Applications

- Connector Endface Inspection
- Patch Panel Cleaning Verification
- Transceiver Port Checking

## Technical Specifications

| Parameter / Specification | Value / Detail          |
|---------------------------|-------------------------|
| Connectivity              | WiFi / USB              |
| Magnification             | 400X                    |
| App Support               | iOS / Android / Windows |

## PASSIVE COMPONENTS

# FanOut Patch Cords

*Small form factor high-density assembly*

PDR MTP/MPO Fanout Patch Cord enables high-density MPO/MTP-to-single-fiber breakout for up to 100G data center applications, available in multiple cable, connector, and fanout configurations.

## Key Features

- Small form factor MPO/MTP breakout to LC, SC, ST, or FC pigtails in 0.9 mm, 2.0 mm simplex, duplex, or Uniboot formats
- Available in 8F, 12F, 24F, 48F, 72F, 96F, and 144F fiber counts; fully custom solutions on request
- Supports single-mode (OS1, OS2) and multimode (OM1, OM2, OM3, OM4) fiber types
- MTP/MPO connectors available in Standard and Elite (low-loss) versions for flexible power budget management
- Both one-end and both-end fanout configurations for equipment-to-panel and panel-to-panel applications
- Micro cable construction in single and double jacket options for compact, high-density routing

## Applications

- Data center interconnections between active equipment, patch panels, and fiber distribution systems
- High-density 40G and 100G structured cabling architectures requiring MPO-to-LC/SC breakouts
- Telecommunications and datacom network backbone and horizontal cabling
- Optical fiber access networks including FTTx and PON deployments

Technical Specifications

| Parameter / Specification                 | Value / Detail                                                                          |
|-------------------------------------------|-----------------------------------------------------------------------------------------|
| MPO Insertion Loss, Typical               | SM Standard: 0.35 dB   SM Low Loss: 0.20 dB MM Standard: 0.35 dB   MM Low Loss: 0.20 dB |
| MPO Insertion Loss, Maximum               | SM Standard: 0.75 dB   SM Low Loss: 0.35 dB MM Standard: 0.60 dB   MM Low Loss: 0.35 dB |
| LC / SC / ST / FC Insertion Loss, Typical | SM Standard: 0.3 dB   SM Low Loss: 0.2 dB MM Standard: 0.3 dB   MM Low Loss: 0.2 dB     |
| Return Loss                               | ≥60 dB (APC)   ≥20 dB (PC)   ≥50 dB (UPC)                                               |
| Durability                                | < 0.2 dB change after 200 insertions                                                    |
| Operating Temperature                     | -40°C to +75°C                                                                          |
| Fiber Mode, Single-Mode                   | OS1, OS2                                                                                |
| Fiber Mode, Multimode                     | OM1, OM2, OM3, OM4                                                                      |
| MPO Connector Type                        | Male or Female; PC or APC end face                                                      |
| Single Ferrule Connector Types            | LC, SC, ST, FC; PC or APC end face                                                      |
| Available Ferrule Counts                  | 4F, 8F, 12F, 24F                                                                        |
| Fiber Count Options                       | 8F, 12F, 24F, 48F, 72F, 96F, 144F; custom available                                     |
| Cable Jacket Diameter                     | 0.9 mm   2.0 mm   3.0 mm   4.5 mm   7.0 mm   8.0 mm                                     |
| Cable Construction                        | Micro single jacket   Micro double jacket                                               |

## CABLE MANAGEMENT DEVICES

# Fiber Distribution Box (FDB)

*Fiber distribution box for splitting and distributing fiber in access networks.*

PDR-FDB24B Fiber Distribution Box is a compact, rugged FTTx termination enclosure supporting straight-joint, branch, and pass-through connectivity with up to 24 SC adaptors, two 1×8 LGX splitters, and separate cable routing for easy maintenance and fault identification.

## Key Features

- Accommodates 1×4, 1×8, 1×16, 2×4, 2×8, and 2×16 PLC steel tube or LGX splitter modules
- Supports wall mounting and pole mounting for flexible field deployment
- Individual link testing possible without disturbing other subscriber connections
- IP65 protection class, suitable for both indoor and outdoor environments
- 24 drop-cable exit ports with dedicated separation of drop cables and fiber loose tubes
- Safe-shaped lock with front-access opening angle exceeding 120 degrees for easy maintenance

## Applications

- FTTx (FTTH / FTTB / FTTC) last-mile access network splitter distribution points
- Pole-mounted and wall-mounted fiber distribution in urban and rural deployments
- Telecom carrier and broadband service provider outdoor cabinet installations
- Multi-dwelling unit (MDU) fiber termination and subscriber connectivity hubs

## Technical Specifications

| Parameter / Specification     | Value / Detail                                               |
|-------------------------------|--------------------------------------------------------------|
| Model                         | PDR-FDB24B                                                   |
| Dimensions (L x W x D)        | 320 × 275 × 105 mm                                           |
| Weight                        | 1.36 kg                                                      |
| Material                      | High Quality Engineering Plastic                             |
| Protection Class              | IP65                                                         |
| Mounting Options              | Wall Mount / Pole Mount                                      |
| Opening Angle                 | > 120 degrees                                                |
| Cable Diameters Supported     | 4 - 7 mm drop cable                                          |
| Total Optical Splice Capacity | 24                                                           |
| Maximum Cable Entries         | 2                                                            |
| Mid-Span Port                 | 1 feeder cable entry (2 ports), 1 entrance port + 1 reserved |
| Drop Cable Exit Ports         | 24 ports                                                     |
| Branching & Distribution      | 2 in, 24 out                                                 |
| LGX Splitter Modules          | 2 x (1×8)                                                    |

## PASSIVE COMPONENTS

# G652D Fiber Spool / Fiber Drum

*Precision-wound G.652D single-mode fiber spool for long-haul, FTTH and data infrastructure.*

Precision-wound ITU-T G.652D single-mode reels with Germania-doped silica core, ultra-low attenuation under 0.200 dB/km at 1550 nm across 1310–1625 nm, individually tested and barcoded for traceability.

## Key Features

- ITU-T G.652D compliant, meets and exceeds international single-mode fiber standards
- Ultra-low attenuation under 0.200 dB/km at 1550 nm for long-haul signal integrity
- Wide operating wavelength range, optimized from 1310 nm to 1625 nm
- Dual-layer UV-curable resin coating for mechanical and environmental protection
- Germania-doped silica core with pure silica cladding
- Precision-wound reels, individually tested and labelled with unique ID and barcode

## Applications

- Long-haul telecommunications backbone
- FTTH and access network deployment
- High-bandwidth data-centre infrastructure
- Spool stock for splicing, testing and field restoration

# Technical Specifications

| Parameter / Specification | Value / Detail                                 |
|---------------------------|------------------------------------------------|
| Fiber Type                | Single Mode, ITU-T G.652D                      |
| Attenuation               | ≤ 0.36 dB/km (1310 nm); ≤ 0.23 dB/km (1550 nm) |
| Cable Cutoff Wavelength   | under 1260 nm                                  |
| Mode Field Diameter       | 9.2 ±0.4 μm at 1310 nm                         |
| Cladding Diameter         | 125 ±0.7 μm                                    |

## PASSIVE COMPONENTS

# Fiber Optic Connector, Field Installable

*Field-installable fast connectors for quick, tool-light on-site termination.*

Field-installable SC connector for tool-light on-site termination with no adhesives or polishing, fully zirconia-ceramic ferrule,  $\leq 0.3$  dB IL,  $\geq 45$  dB (PC)/ $\geq 55$  dB (APC) RL, re-assemblable.

## Key Features

- Field installable without adhesives, polishing, or electrical power, ready for immediate on-site use
- Ferrule end-face geometry compliant with Telcordia GR-326 interferometric measurement standard
- Re-assemblable design reduces connection defect rate and simplifies field maintenance and rework
- Fully zirconia-ceramic ferrule: nominal 2.5 mm diameter,  $\pm 0.5$  mm tolerance,  $\leq 1$   $\mu$ m fiber concentricity
- Low insertion loss  $\leq 0.3$  dB and high return loss  $\geq 45$  dB (PC) /  $\geq 55$  dB (APC)
- Type-A: compact  $52.4 \times 9 \times 7.3$  mm body, compatible with  $3.1 \times 2$  mm flat cable

## Applications

- FTTx last-mile fiber drops and subscriber premises installations requiring rapid field termination
- Enterprise and campus network fiber installations demanding fast, reliable connector assembly
- Telecommunications access network buildouts where on-site polishing equipment is impractical
- Field restoration and emergency repair of optical fiber links with minimal downtime

# Technical Specifications

| Parameter / Specification | Value / Detail                              |
|---------------------------|---------------------------------------------|
| Connector Interface       | SC                                          |
| Polish Type               | PC / APC                                    |
| Body Type                 | Type-A / Type-B                             |
| Ferrule Material          | Fully Zirconia-Ceramic                      |
| Insertion Loss            | ≤0.3 dB                                     |
| Return Loss               | ≥45 dB (PC) / ≥55 dB (APC)                  |
| Compatible Cable          | 3.1×2 mm flat / 3 mm drop / 2.0 mm / 0.9 mm |

## PASSIVE COMPONENTS

# Fiber Optic Patch Cords & Pigtails

*Broadest choice of patchcords in the industry*

PDR Fiber Optic Patch Cords and Pigtails offer IEC/EIA/TIA/GR-326 compliant, factory-terminated singlemode and multimode assemblies with multiple connector, cable, and fiber options.

## Key Features

- Widest connector selection: FC, DIN, LC, MU, MTRJ, SC, ST, MPO, D4, Biconic, SMA, E-2000
- Simplex, duplex, and fanout (up to 24F) configurations; 0.9 mm, 2.0 mm, and 3.0 mm cable diameters
- Exchangeability and repeatability both  $\leq 0.2$  dB for consistent mating performance
- Singlemode fibers: OS2, G.657A1, G.657A2, multimode fibers: OM1, OM2, OM3, OM4
- Low insertion loss  $\leq 0.3$  dB (PC) for both SM and MM; high return loss  $\geq 50$  dB UPC,  $\geq 60$  dB APC
- Boot styles: standard, short boot, 90° bendable boot, and uniboot for high-density applications

## Applications

- Data centre structured cabling and patch panel interconnections in high-density SM and MM environments
- Enterprise and campus LAN backbone and horizontal cabling with SM (OS2/G.657) or MM (OM3/OM4) fiber
- Telecommunications central office and outside plant cross-connects requiring low-loss, reliable patch cords
- FTTx access network distribution point and subscriber premises connections with bend-insensitive G.657A fiber

## Technical Specifications

| Parameter / Specification | Value / Detail                                                          |
|---------------------------|-------------------------------------------------------------------------|
| Cable Diameter            | 0.9 mm   2.0 mm   3.0 mm (or custom)                                    |
| Cable Configuration       | Simplex   Duplex   Fanout (up to 24F)                                   |
| SM Fiber Type             | OS2   G.657A1   G.657A2                                                 |
| MM Fiber Type             | OM1   OM2   OM3   OM4                                                   |
| Connector Types           | FC   LC   SC   ST   MPO   MU   MTRJ   DIN   D4   Biconic   SMA   E-2000 |
| Polish                    | UPC   APC   PC                                                          |
| SM Insertion Loss (PC)    | ≤0.3 dB                                                                 |
| SM Return Loss, UPC       | ≥50 dB                                                                  |
| SM Return Loss, APC       | ≥60 dB                                                                  |
| MM Insertion Loss (PC)    | ≤0.3 dB                                                                 |
| MM Return Loss (PC)       | ≥35 dB                                                                  |
| Exchangeability           | ≤0.2 dB                                                                 |
| Repeatability             | ≤0.2 dB                                                                 |
| Operating Temperature     | -20°C to +75°C                                                          |

## TEST AND MEASUREMENT EQUIPMENT

# Fusion Splicer PDR618H

*Handheld core-alignment fusion splicer with fast splice and heating times.*

The PDR618H is a compact handheld fusion splicer for SM, MM, DS, and NZDS fibers, offering fast splicing, quick heat shrinking, and automatic ARC calibration for reliable field and lab performance.

## Key Features

- Handheld, compact and lightweight design optimised for field and indoor deployment
- 5200 mAh hot-swappable lithium-ion battery, typical 200+ splice/heat cycles per charge
- USB 2.0 port for software updates and splice data export
- Ultra-fast 8-second splicing time and 26-second tube heating time
- Automatic ARC calibration via real-time temperature and air pressure sensing
- Dual CMOS camera system with 4.3-inch colour LCD display

## Applications

- Fiber-to-the-Home (FTTH) and Fiber-to-the-Building (FTTB) network installations
- Cable TV (CATV) and broadband infrastructure deployment
- Telecommunications network construction and maintenance
- Data centre interconnects and enterprise LAN cabling

## Technical Specifications

| Parameter / Specification | Value / Detail                                                                          |
|---------------------------|-----------------------------------------------------------------------------------------|
| Fiber Count               | Single                                                                                  |
| Cladding Diameter         | 80 to 150 µm                                                                            |
| Coating Diameter          | 160 to 900 µm                                                                           |
| Typical Splice Loss       | SM: 0.02 dB   MM: 0.01 dB   DS: 0.04 dB NZDS: 0.04 dB   G.657: 0.02 dB (ITU-T Standard) |
| Return Loss               | > 60 dB                                                                                 |
| Fiber Cleaved Length      | 10 to 16 mm (coating dia. < 250 µm) 16 mm (coating dia. 250 to 900 µm)                  |
| Splicing Programs         | 40 groups                                                                               |
| Operating Mode            | Manual / Automatic                                                                      |
| Auto-Heating              | Available                                                                               |
| Typical Splicing Time     | 8 seconds                                                                               |
| Tube Heating Time         | 26 seconds (60 mm and 40 mm shrinkable sleeves)                                         |
| Fiber View Magnification  | 250× (X or Y axis)   125× (X and Y axis)                                                |
| Viewing & Display         | 2× CMOS cameras, 4.3-inch colour LCD monitor                                            |
| Splice Result Storage     | 4000 results                                                                            |

## CABLE MANAGEMENT DEVICES

# Heat Shrink Splice Closure

*Sealed splice closure protecting fiber joints in outdoor and buried networks.*

The PDR Fiber Optic Splice Closure is an IP68 weatherproof enclosure with heat shrink or re-enterable options, supporting up to 6 cable entries and 144 splices for outdoor fiber optic networks.

## Key Features

- IP 68 sealing protection, suitable for fully immersed and buried deployments
- Unique strength member clamp assembly prevents cable sheath movement with temperature changes
- Resistant to UV radiation and temperature fluctuation for deployment in all environments
- Zero additional insertion loss when fibers are routed through splice trays
- No specialized tools required, minimizes installation time and training costs
- Bend controls in splicing tray guide and protect pigtail cables, ensuring controlled bending radius

## Applications

- Fiber-to-the-Home (FTTH) and Fiber-to-the-Building (FTTB) outdoor splice points
- Underground duct and direct-buried fiber optic cable joint protection
- Aerial fiber optic cable splicing on poles and overhead installations
- Telecommunications network straight-through and branching joint applications

## Technical Specifications

| Parameter / Specification | Value / Detail                                                             |
|---------------------------|----------------------------------------------------------------------------|
| Dimension (L × D)         | 530 mm × 210 mm                                                            |
| Weight                    | < 2.7 kg                                                                   |
| Type                      | Butt type                                                                  |
| Cable Sealing             | Heat shrink sleeves (Heat Shrink Type)   Mechanical seal (Mechanical Type) |
| Dome Material             | PPCP                                                                       |

## CABLE MANAGEMENT DEVICES

# Horizontal Splice Closure

*Horizontal splice closure for inline protection of fiber splices.*

Re-enterable IP65 outdoor closure for duct, aerial, and wall mounting, supporting up to 48 splices (2×24) in a UV-resistant ribbed 445×240×140 mm body.

## Key Features

- Suitable for cut, uncut, and taut sheath fiber optic cable applications
- Integrated sheath retention and central strength member fastening
- High resistance to chemical attack and UV degradation for extended outdoor life
- Compatible with all standard fiber optic cable types and diameters
- Ribbed closure construction using UV-resistant, lightweight plastic
- Supports up to 48 splices (2 × 24) with 16 SC simplex adapter capacity

## Applications

- Fiber-to-the-Home (FTTH) and Fiber-to-the-Building (FTTB) feeder cable splicing
- Duct, aerial, and wall-mounted fiber optic cable distribution installations
- Telecommunications network outdoor splice points and cable junction management
- Cable TV (CATV) and broadband infrastructure outdoor splice enclosures

# Technical Specifications

| Parameter / Specification | Value / Detail                  |
|---------------------------|---------------------------------|
| Model                     | HTSC-TL16                       |
| Dimensions (L×W×H)        | 445 × 240 × 140 mm              |
| Splice Capacity           | 48 splices (2 × 24)             |
| Main Cable Ports          | 4 ports, max 12 mm diameter     |
| Drop Cable Ports          | 16 ports, 3 mm / 3×2 drop cable |
| Mounting Options          | Duct, Aerial, Wall              |
| Protection Rating         | IP65                            |

## PASSIVE COMPONENTS

# Hybrid Adapter

*Hybrid fiber optic adapters that mate different connector types.*

Mates two dissimilar connector types (SC-FC, LC-SC, ST-FC, etc.) via a precision ceramic/phosphor-bronze sleeve,  $\leq 0.3$  dB IL and  $\geq 50$  dB RL, for SM/MM fiber in simplex or duplex.

## Key Features

- Enables connection between different connector types, SC-FC, LC-SC, LC-FC, ST-FC, LC-ST, and more
- Low insertion loss  $\leq 0.3$  dB and high return loss  $\geq 50$  dB (single-mode)
- Available in simplex and duplex configurations to suit varied cabling requirements
- High-precision ceramic or phosphor bronze alignment sleeve for superior optical performance
- Compatible with both single-mode (SM) and multi-mode (MM) fibers

## Applications

- Fiber-to-the-Home (FTTH) and Fiber-to-the-Building (FTTB) network installations
- Cable TV (CATV) and broadband infrastructure interconnects
- Telecommunications network upgrades requiring connector-type migration
- Data center cross-connects and enterprise LAN cabling

## Technical Specifications

| Parameter / Specification | Value / Detail                                   |
|---------------------------|--------------------------------------------------|
| Adapter Type              | Hybrid (SC-FC, SC-LC, LC-ST, ST-FC, LC-FC, etc.) |
| Fiber Type                | Single-mode / Multi-mode                         |
| Insertion Loss            | ≤ 0.3 dB                                         |
| Return Loss               | ≥ 50 dB (SM)                                     |
| Operating Wavelength      | 1310 nm / 1550 nm                                |
| Durability                | ≥ 1000 mating cycles                             |

## PASSIVE COMPONENTS

# Mode Conditioning Patchcord

*Gigabit launch patchcord that runs single-mode laser sources over legacy multimode fiber without DMD.*

Offset-launch patchcord that runs single-mode LX/LH laser sources over legacy 62.5/125 or 50/125 multimode fiber, eliminating Differential Mode Delay for reliable Gigabit Ethernet.

## Key Features

- Ensures Gigabit transmission over multimode fiber
- Offset launch technique from single-mode to multimode
- Available with various connector types
- High durability for repeated matings ( $\leq 0.20$  dB typical change after 500 matings)
- Eliminates Differential Mode Delay (DMD)
- Compatible with 62.5/125  $\mu\text{m}$  or 50/125  $\mu\text{m}$  MMF cable plant
- High return loss and low insertion loss

## Applications

- Gigabit Ethernet (1000BASE-LX / LH) over legacy multimode fiber
- Campus and building backbone links reusing existing multimode infrastructure
- GBIC / SFP long-wavelength transceiver deployments on MMF cable plant
- Data centre and enterprise LAN interconnects

## Technical Specifications

| Parameter / Specification | Value / Detail                           |
|---------------------------|------------------------------------------|
| Cable Plant               | 62.5/125 or 50/125 Multimode Fiber (MMF) |
| Insertion Loss (MMF–SMF)  | ≤ 2.1 dB (50/125); ≤ 2.6 dB (62.5/125)   |
| Return Loss               | ≥ 50 dB (UPC); ≥ 65 dB (APC)             |
| Durability                | ≤ 0.20 dB typical after 500 matings      |
| Operating Temperature     | –20 °C to +75 °C                         |

## PASSIVE COMPONENTS

# MPO Cable Assembly

*High-density multi-fiber connector system*

PDR MPO Cable Assemblies are high-density 4–24 fiber OS2/OM3/OM4/OM5 trunk cables with MT ferrules, ribbon or round cables, and male/female PC/APC connectors for fast data center deployment.

## Key Features

- High-density MT ferrule with precision guide pins for reliable multi-fiber alignment in 4, 8, 12, and 24F configurations
- Cable types: jacketed ribbon, bare ribbon, and round cable in single and double jacket constructions
- Male and female connector options with push-pull latching for tool-free, secure mating
- Fiber options: OS2 (singlemode), OM3, OM4, OM5 (multimode), supports 10GbE, 40GbE, and 100GbE migration
- End-face polish: PC and APC; connector housings available in Green, Yellow, Beige, and Aqua colours
- SM insertion loss: 0.35 dB typical (standard), 0.20 dB typical (low loss); maximum 0.75 dB / 0.35 dB

## Applications

- Data centre backbone and inter-rack cabling for 10GbE, 40GbE, and 100GbE high-density architectures
- Telecom and datacom network infrastructure requiring rapid, high-density multi-fiber connectivity
- MPO trunk cable distribution in structured cabling systems between main and horizontal distribution areas
- Optical fiber access networks and central office patch fields with large fiber counts

Technical Specifications

| Parameter / Specification                     | Value / Detail                                        |
|-----------------------------------------------|-------------------------------------------------------|
| Fiber Mode                                    | SM: OS2   MM: OM3, OM4, OM5                           |
| Fiber Count                                   | 4F   8F   12F   24F   48F   72F   96F   144F   Custom |
| Cable Type                                    | Jacketed Ribbon   Bare Ribbon   Round                 |
| Cable Material                                | LSZH   PVC                                            |
| Cable Diameter, Single Jacket 12F             | 3.0 mm                                                |
| Cable Diameter, Double Jacket 12F/24F         | 4.5 mm                                                |
| Tensile Strength, Single 12F (short/long)     | 200 N / 100 N                                         |
| Tensile Strength, Double 12F/24F (short/long) | 400 N / 150 N                                         |
| Crush Resistance, Single 12F                  | 500 N/100 mm                                          |
| Crush Resistance, Double 12F                  | 1,000 N/100 mm                                        |
| Min. Bending Radius                           | 10×D (installed)   20×D (under load)                  |
| Cable Operating Temperature                   | , 20°C to +70°C                                       |
| MPO Connector Type                            | Male or Female; PC or APC end face                    |
| Available Ferrule Counts                      | 4, 8, 12, 24                                          |

## PASSIVE COMPONENTS

# Fiber Optic Cleaner Pen MPO

*One-click cleaner pen for multi-fiber MPO/MTP connector end-faces.*

PDR MPO/MTP Fiber Optic Cleaner Pen provides fast, dry cleaning of male and female MPO/MTP connectors, cleaning all 12 fibers in one push, fitting adapters and exposed ferrules, and delivering up to 500 cleaning cycles.

## Key Features

- Pen-style handheld design with narrow profile for tightly-spaced MPO adapter panels
- Dry mechanical cleaning, no alcohol or wet chemicals required
- Effective removal of dust, oils, and particulate contaminants per IEC 61300-3-35
- Cleans all 12 fiber end-faces simultaneously in a single push-and-release action
- Compatible with male (with guide pins) and female (without guide pins) MPO/MTP ferrules
- Intermatability with FOCIS-5 (IEC 61754-7) compliant MPO connectors

## Applications

- Data centre MPO/MTP patch panel and trunk cable maintenance
- Telecommunications network construction, testing, and field maintenance
- Fiber-to-the-Home (FTTH) and Fiber-to-the-Building (FTTB) network installations
- Cable TV (CATV) and broadband infrastructure connector cleaning

Technical Specifications

| Parameter / Specification | Value / Detail                                          |
|---------------------------|---------------------------------------------------------|
| Compatible Connectors     | MPO / MTP                                               |
| Connector Standard        | FOCIS-5 (IEC 61754-7)                                   |
| Fiber Mode                | Single-mode (SM) and Multimode (MM)                     |
| Ferrule Configuration     | Male (with guide pins) and Female (without guide pins)  |
| Polish Type               | PC (Physical Contact) and APC (Angled Physical Contact) |
| Cleaning Method           | Dry mechanical, no alcohol or solvents required         |
| Fibers Cleaned per Stroke | 12 fibers simultaneously                                |
| Cleaning Capacity         | Up to 500 cleanings per unit                            |
| Operation                 | One-handed push-and-release mechanism                   |
| Cleaning Compatibility    | Adapter-mounted connectors and exposed jumper ferrules  |
| Contaminant Types         | Dust, oils, fingerprints, and particulate matter        |
| Cleaning Standard         | IEC 61300-3-35 (fiber end-face cleanliness)             |
| Design                    | Pen-style, handheld, narrow body for panel access       |

## TEST AND MEASUREMENT EQUIPMENT

# Next Gen Optical Fusion Splicer (OPM + VFL)

*Next-generation fusion splicer with integrated power meter and visual fault locator.*

The PDR4107S is a compact core-alignment fusion splicer for SM, MM, DS, and NZDS fibers, featuring 0.02 dB splice loss, a 5-second splice time, and built-in OPM and VFL for all-in-one field splicing.

## Key Features

- Compact, lightweight design ideal for field installation and network maintenance
- Fast 5-second typical splicing time with automatic and manual operating modes
- Built-in Optical Power Meter (OPM), measurement range -50 to +26 dBm at 10 wavelengths
- Core alignment technology ensuring high-precision, low-loss fiber splicing
- Supports SM, MM, DS, and NZDS fiber types for maximum versatility
- Built-in Visual Fault Locator (VFL) with 10 mW output for fault identification

## Applications

- Fiber-to-the-Home (FTTH) and Fiber-to-the-Building (FTTB) network installations
- Cable TV (CATV) and broadband infrastructure rollout
- Telecommunications network construction, deployment, and maintenance
- Data centre interconnects and enterprise LAN fiber cabling

Technical Specifications

| Parameter / Specification | Value / Detail                                          |
|---------------------------|---------------------------------------------------------|
| Applicable Fibers         | SM   MM   DS   NZDS                                     |
| Fiber Count               | Single                                                  |
| Cladding Diameter         | 80 to 150 μm                                            |
| Coating Diameter          | 1 to 3 mm                                               |
| Fiber Aligning Method     | Automatic / Manual                                      |
| Focusing Method           | Four-motor automatic focusing                           |
| Typical Splice Loss       | SM: 0.02 dB   MM: 0.01 dB   DS: 0.04 dB   NZDS: 0.04 dB |
| Return Loss               | > 60 dB                                                 |
| Splicing Programs         | 53 preset + 40 user-defined                             |
| Splicing Modes            | Auto / Calibrate / Normal / Special                     |
| Operating Mode            | Automatic / Manual                                      |
| Typical Splicing Time     | 5 seconds (standard SM fiber)                           |
| Tube Heating Programs     | 9 preset + 24 user-defined                              |
| Tube Heating Time         | 15 seconds   Protection sleeve: 60 mm / 40 mm           |

## ACTIVE COMPONENTS &gt; OPTICAL LINE PROTECTION SYSTEM (OLPS)

# Optical Line Protection System (OLPS)

*1U optical line protection with automatic switchover to a backup fiber path on failure.*

PDR Optical Line Protection System (OLP): Compact 1U optical-layer protection with real-time monitoring, automatic fiber switching, 1+1/1:1/1-1 protection, redundant AC/DC power, and NMS support.

## Key Features

- Compact 1U, 19-inch rack-mountable form factor
- Real-time optical power monitoring of both channels
- Configurable switching delay (0 to 999 s) and auto switchback
- Automatic fiber line switching on failure or degradation
- Three protection architectures: 1+1, 1:1 and 1-1
- Front-panel LCD with key control

## Applications

- Telecom transmission network path redundancy
- Emergency rerouting of live traffic
- Real-time fiber quality surveillance
- FTTH/FTTB and access-network resilience

Technical Specifications

| Parameter / Specification | Value / Detail                     |
|---------------------------|------------------------------------|
| Protection Architecture   | 1+1   1:1   1-1                    |
| Form Factor               | 1U rack-mountable (19-inch)        |
| Working Wavelength        | 1310 nm or 1550 nm (selectable)    |
| Switching Threshold       | Default -30 dBm (configurable)     |
| Switching Delay           | Immediate or 1 to 999 s            |
| Auto Switchback           | Configurable 1 to 999 minutes      |
| Management                | RJ-45 Ethernet   RS-232 DB9        |
| Power Supply              | 220V AC   -48V DC (dual redundant) |

## PASSIVE COMPONENTS

# PLC Splitter

*Planar lightwave circuit splitters for low-loss PON power splitting.*

PDR PLC Splitters offer passive optical power splitting over 1260–1650 nm in 1×N/2×N configurations with multiple packaging options for FTTx, PON, LAN, CATV, and test applications.

## Key Features

- Planar waveguide chip technology for low insertion loss, high uniformity, and excellent long-term stability
- 1×N configurations: 1×2, 1×4, 1×8, 1×16, 1×32, 1×64
- 2×N configurations: 2×2, 2×4, 2×8, 2×16, 2×32, 2×64
- Wide operating wavelength: 1260 to 1650 nm, covers all standard PON and WDM bands
- Low polarization-dependent loss (PDL): ≤0.2 dB (1×2) to ≤0.4 dB (1×64)
- High directivity ≥55 dB and return loss ≥55 dB (1×N) / ≥50 dB (2×N)

## Applications

- FTTx (FTTP, FTTH, FTTN, FTTC) passive optical network (PON) power splitting at distribution points
- GPON, XGS-PON, and EPON access network subscriber branching and signal distribution
- Local Area Network (LAN) fiber distribution in enterprise, campus, and data centre environments
- CATV hybrid fiber-coax (HFC) network signal splitting and distribution Technical Specifications (label + value pairs)

Technical Specifications

| Parameter / Specification                 | Value / Detail                         |
|-------------------------------------------|----------------------------------------|
| Port Configurations                       | 1×2   1×4   1×8   1×16   1×32   1×64   |
| Fiber Type                                | SMF-28e (default)   customer specified |
| Operating Wavelength                      | 1260 to 1650 nm                        |
| Insertion Loss, 1×2 / 1×4 / 1×8           | 4.0 dB / 7.4 dB / 10.5 dB              |
| Insertion Loss, 1×16 / 1×32 / 1×64        | 13.6 dB / 16.9 dB / 21.0 dB            |
| Loss Uniformity (max), 1×2 / 1×4 / 1×8    | 0.5 dB / 0.7 dB / 1.0 dB               |
| Loss Uniformity (max), 1×16 / 1×32 / 1×64 | 1.4 dB / 1.8 dB / 2.0 dB               |
| Return Loss (min), 1×N                    | ≥55 dB                                 |
| PDL (max), 1×2 to 1×8 / 1×16 to 1×64      | ≤0.2 to 0.3 dB / ≤0.3 to 0.4 dB        |
| Directivity (min)                         | ≥55 dB                                 |
| Wavelength-Dependent Loss (WDL, max)      | 0.3 to 0.8 dB (varies by port count)   |
| Temperature-Dependent Loss (TDL, max)     | 0.5 to 1.0 dB (varies by port count)   |
| 2×N Return Loss (min)                     | ≥50 dB                                 |
| Operating Temperature                     | −40°C to +85°C                         |

## TEST AND MEASUREMENT EQUIPMENT

# Pocket OTDR

*Ultra-compact Nano OTDR with built-in VFL and power meter for quick field testing.*

The PDR Nano OTDR is a compact, smartphone-connected dual-wavelength (1310/1550 nm) OTDR for precise fiber testing, loss analysis, and real-time measurements via the Smart-OTDR app.

## Key Features

- Ultra-compact nano size, fits in a shirt pocket for true field portability
- Dual-wavelength operation at 1310 nm and 1550 nm for comprehensive fiber testing
- Wireless Bluetooth / Wi-Fi connectivity to the Smart-OTDR smartphone app (Android & iOS)
- Real-time OTDR trace display, zoom, and cursor analysis on smartphone screen
- Measures Total Loss, Total ORL, Average Loss, and Distance per event automatically
- Multiple pulse widths for optimised dead-zone and dynamic-range trade-off

## Applications

- Fiber-to-the-Home (FTTH) and Fiber-to-the-Building (FTTB) acceptance testing
- Telecommunications network construction, commissioning, and maintenance
- Cable TV (CATV) and broadband passive optical network (PON) verification
- Data centre interconnect fiber certification and fault localisation

Technical Specifications

| Parameter / Specification | Value / Detail                                |
|---------------------------|-----------------------------------------------|
| Fiber Type                | Single-Mode (ITU-T G.652 / G.657)             |
| Wavelengths               | 1310 nm   1550 nm                             |
| Dynamic Range             | 26 dB @ 1310 nm   24 dB @ 1550 nm             |
| Distance Range            | 0.1 km, 60 km (auto & manual)                 |
| Distance Resolution       | 0.01 m                                        |
| Distance Accuracy         | ±1 m + ±0.005% × distance                     |
| Event Dead Zone           | ≤ 1 m (typical, with 10 ns pulse)             |
| Attenuation Dead Zone     | ≤ 4 m (typical, with 10 ns pulse)             |
| Pulse Widths              | 10 ns / 30 ns / 100 ns / 300 ns / 1 μs / 3 μs |
| Sampling Resolution       | 4 cm                                          |
| Loss Threshold            | 0.01 dB resolution                            |
| Return Loss (ORL)         | Measured in dB, displayed in app              |
| Averaging Time            | 5 s / 15 s / 30 s / 60 s / 180 s              |
| Connectivity              | Bluetooth 4.0   Wi-Fi 802.11 b/g/n            |

## PASSIVE COMPONENTS

# POF — Plastic Optical Fiber Patchcord

*Large PMMA core for easy alignment*

PDR Plastic Optical Fiber (POF) Cable Assemblies provide cost-effective, EMI-immune connectivity with SI-POF cables, LC/RedLink™/VersaLink® patch cords, reel options for custom installation, UL VW-1 jackets, and compatibility with standard POF transceivers.

## Key Features

- 1 mm step-index SI-POF with numerical aperture 0.5, compliant with IEC 60793-2-40 Class A4a
- Simplex cable OD 2.2 mm; duplex (figure-of-8) minor axis 2.2 mm, major axis 4.4 mm
- Minimum bend radius 25 mm; repeat bending endurance  $\geq 10,000$  cycles ( $\leq 1$  dB additional loss)
- Low attenuation: 0.19 dB/m at 650 nm, ideal for short-range links up to tens of metres
- UL VW-1 flame-retardant jacket option available for building and equipment installations
- Tensile rating: 70 N simplex / 140 N duplex cable

## Applications

- Control links within high-voltage electrical control equipment requiring full galvanic isolation
- Industrial automation links between equipment requiring maintained electrical isolation
- Data communications systems in environments with extreme electromagnetic interference (EMI)
- Rugged short-range optical links in harsh factory, process control, and automotive environments

## Technical Specifications

| Parameter / Specification         | Value / Detail                                              |
|-----------------------------------|-------------------------------------------------------------|
| Fiber Standard                    | IEC 60793-2-40 Class A4a (SI-POF)                           |
| Numerical Aperture                | 0.5                                                         |
| Attenuation at 650 nm             | ≤0.19 dB/m                                                  |
| Fiber Diameter (incl. cladding)   | 0.94 to 1.06 mm (nominal 1.0 mm)                            |
| Cable Diameter, Simplex           | 2.13 to 2.27 mm (nominal 2.2 mm)                            |
| Cable Diameter, Duplex Minor Axis | 2.13 to 2.27 mm                                             |
| Cable Diameter, Duplex Major Axis | 4.3 to 4.5 mm                                               |
| Jacket Material                   | Flame-resistant PE or PVC                                   |
| Jacket Color                      | Black; Duplex cable has white identifying bar on one strand |
| UL Flame Rating                   | VW-1                                                        |
| Tensile Stress, Simplex / Duplex  | 70 N / 140 N                                                |
| Minimum Bend Radius               | 25 mm                                                       |
| Repeat Bending Endurance          | ≥10,000 cycles (≤1 dB additional loss)                      |
| Operating Temperature             | -40°C to +85°C                                              |

## TEST AND MEASUREMENT EQUIPMENT

# PON Power Meter

*PON power meter with pass-through measurement for live FTTx networks.*

The PDR4213B is a handheld PON Power Meter for simultaneous 1310/1490/1550 nm measurement, providing fast, accurate FTTH/GPON/EPON testing for installation and maintenance.

## Key Features

- Simultaneous three-wavelength display: 1310 nm, 1490 nm, and 1550 nm on a single screen
- High accuracy of  $\pm 0.25$  dB for reliable installation and maintenance testing
- Dedicated SC/APC and SC/UPC optical ports for direct connection to PON splitters
- Wide power measurement range of -50 dBm to +26 dBm covers all PON signal levels
- Large backlit LCD for clear readings in bright sunlight or low-light conditions
- USB port for data logging, PC connectivity, and firmware upgrades

## Applications

- FTTH / FTTB installation acceptance testing and power budget verification
- Simultaneous upstream (1310 nm) and downstream (1490 / 1550 nm) signal verification
- GPON and EPON network commissioning, maintenance, and fault localisation
- PON splitter loss measurement and passive component characterisation

Technical Specifications

| Parameter / Specification | Value / Detail                               |
|---------------------------|----------------------------------------------|
| Wavelengths               | 1310 nm   1490 nm   1550 nm                  |
| Measurement Mode          | Simultaneous 3-channel power measurement     |
| Power Range               | -50 dBm to +26 dBm                           |
| Resolution                | 0.01 dB   0.001 mW                           |
| Accuracy                  | +/- 0.25 dB (at reference wavelength, 23 C)  |
| Display Units             | dBm   dBr   mW (user selectable)             |
| Detector Type             | InGaAs photodiode                            |
| Optical Connector         | SC/APC (SC/UPC adapter included)             |
| Connector Interface       | SC adapter port with 2.5 mm ferrule          |
| Data Storage              | Up to 99 measurement records (SAVE function) |
| Data Interface            | USB 2.0 (data transfer and firmware upgrade) |
| Battery                   | 3 x AA alkaline batteries                    |
| Battery Life              | >= 40 hours continuous operation             |
| Auto Power-Off            | Configurable: 5 / 10 / 15 min or disabled    |

## CABLE MANAGEMENT DEVICES

# Rack Mount Fiber Management System

*Modular 19-inch rack-mount fiber management system for high-density termination.*

PDR 19-inch Rack Mount Fiber Management System provides modular, scalable, high-density fiber management in durable 1U/2U/4U CRCA steel enclosures with sliding tray access, supporting 6–192 fiber terminations.

## Key Features

- 19-inch rack mount enclosure available in 1U, 2U, and 4U heights for scalable fiber density
- Removable top cover and sliding tray mechanism for quick and easy access to service connections
- Slide-out mechanism provides excellent cable management and organizes cable slack efficiently
- Supports 6F to 192F fiber terminations per enclosure, FC, SC, ST, and LC connector types
- Multiple cable entry and exit openings on all models for flexible routing and installation
- Maintains minimum bend radius of pigtails and patch cords to protect fiber integrity

## Applications

- Data centre main distribution frames (MDF) and structured cabling infrastructure
- Enterprise LAN fiber backbone and horizontal cabling patch points
- Telecommunications exchange and central office fiber termination panels
- FTTH and FTTB aggregation points in multi-dwelling unit (MDU) installations

Technical Specifications

| Parameter / Specification | Value / Detail                                   |
|---------------------------|--------------------------------------------------|
| Rack Compatibility        | 19-inch standard rack mount                      |
| Available Rack Heights    | 1U   2U   4U                                     |
| Fiber Capacity Range      | 6F to 192F per enclosure                         |
| Connector Types           | FC / SC / ST (XX series)   LC / PC (LC series)   |
| Material                  | CRCA (Cold-Rolled Close-Annealed steel)          |
| Finish                    | Structure finish and powder coated, 80 to 100 µm |
| Colour                    | RAL 7032 (standard)   Customised on request      |
| Cable Entry               | Rear                                             |
| Cable Mounting            | Horizontal                                       |
| Splice Tray               | Hinged type ABS                                  |
| Top Cover                 | Removable for service access                     |
| Access Mechanism          | Sliding tray                                     |
| Rodent Cover              | Available (optional)                             |
| Dimensions (1U models)    | 485 mm × 315 mm × 45 mm                          |

## PASSIVE COMPONENTS

# FTTX Smart Bullet Drop Cable

*FTTXSmart pre-connectorized bullet drop cable for fast last-mile FTTH deployment.*

Single-fiber SC/APC drop assembly with a Ø5.9 mm bullet head for traction pulling through narrow ducts, ≤0.3 dB IL, ≥60 dB RL over 1260–1650 nm, 100% optically tested.

## Key Features

- Bullet-shaped connector head (Ø5.9 mm) with guide-wire hole for traction installation through narrow ducts
- SC/APC connectors with zirconia ceramic ferrule: insertion loss ≤0.3 dB (typ. 0.2 dB), return loss ≥60 dB
- APC ferrule angle  $8^{\circ} \pm 0.3^{\circ}$  for superior back-reflection performance in FTTH and PON networks
- Pull-through design for free and occupied ducts, no large bore holes, conduits, or specialist tools required
- Wide operating wavelength 1260 to 1650 nm covers all standard telecom bands including O, E, S, C, L, and U
- Mechanical durability ≥500 insertion/removal cycles; repeatability and interchangeability ≤0.1 dB

## Applications

- FTTH and FTTx last-mile fiber drops where bullet-head traction installation through narrow ducts is required
- Outdoor and aerial fiber drop deployments requiring UV-resistant, high-tensile cable assemblies
- Micro-conduit and wall-chase installations in residential and commercial MDU premises
- PON and GPON access network subscriber connection points demanding low back-reflection SC/APC terminations

## Technical Specifications

| Parameter / Specification | Value / Detail                                     |
|---------------------------|----------------------------------------------------|
| Connector Type            | SC/APC bullet-shaped (near-end) / SC/APC (far-end) |
| Bullet Connector Diameter | Ø5.9 mm                                            |
| Operating Wavelength      | 1260 to 1650 nm                                    |
| Insertion Loss            | ≤0.3 dB (typ. 0.2 dB)                              |
| Return Loss               | ≥60 dB                                             |
| Tractive Force            | ≥150 N                                             |
| Working Temperature       | −40°C to +70°C                                     |

## TEST AND MEASUREMENT EQUIPMENT &gt; POWER METER

# Mini Optical Power Meter

*Palm-sized handheld optical power meter for FTTH continuity and link-loss testing.*

Palm-sized handheld meter for FTTH continuity and link-loss testing, six calibrated wavelengths (850–1625 nm),  $\pm 0.2$  dB accuracy, 0.01 dB resolution, SM/MM via 2.5 mm connector, up to 100 h battery.

## Key Features

- Palm-sized, pocket-carry handheld design for one-hand field use
- Six calibrated wavelengths: 850 / 1300 / 1310 / 1490 / 1550 / 1625 nm
- Optional visual fault locator (1 mW or 10 mW, CW and glint) and LED SOS output
- Battery autonomy above 100 hours; runs on 2× AA or AC/DC adapter
- Wide measurement range with high 0.01 dB resolution and  $\pm 0.2$  dB accuracy
- Supports single-mode and multimode fiber via 2.5 mm universal connector
- Auto power-off after 10 minutes (cancellable) to conserve battery
- One-year warranty with three-year recommended calibration interval

## Applications

- FTTH in-house installation and continuity testing
- Fiber link transmission quality evaluation
- Field service and troubleshooting by technicians
- Optical fiber network installation and maintenance
- Single-mode and multimode fiber measurement

# Technical Specifications

| Parameter / Specification | Value / Detail                            |
|---------------------------|-------------------------------------------|
| Display Range (PDR60T)    | +6 to -70 dBm (1310/1490/1550/1625 nm)    |
| Display Range (PDR60C)    | +26 to -50 dBm (1310/1490/1550/1625 nm)   |
| Accuracy                  | ±0.2 dB                                   |
| Calibrated Wavelengths    | 850 / 1300 / 1310 / 1490 / 1550 / 1625 nm |
| Display Resolution        | 0.01 dB                                   |
| Battery Life              | Above 100 hours                           |
| Power Supply              | 2× AA or AC/DC adapter                    |
| Dimensions / Weight       | 105 × 52 × 34 mm; ~100 g                  |

## ACTIVE COMPONENTS

# 100G & 100G BIDI

*Full range of MSA-compliant transceivers from 1.25G to 400G across SFP and QSFP form factors.*

PDR optical transceivers deliver hot-pluggable, MSA-compliant 1.25G–400G connectivity with SFP to QSFP-DD form factors, DDM, and SM/MM, CWDM/DWDM/BiDi support.

## Key Features

- MSA-compliant form factors: SFP, SFP+, SFP28, QSFP+, QSFP28, QSFP-DD and Copper SFP
- Single-mode and multimode options with reach up to 80 km
- Digital Diagnostic Monitoring (DDM/DOM) on optical variants
- Aggregate data rates from 1.25 Gbps to 400 Gbps
- Hot-pluggable with duplex LC (SFP) and MPO/MTP (QSFP) receptacles
- CWDM, DWDM and BiDi variants available on request

## Applications

- Telecom backbone, metro Ethernet and long-haul DWDM
- Enterprise and campus LAN uplinks
- Data-centre server-to-switch and spine aggregation

## Technical Specifications

| Parameter / Specification | Value / Detail                        |
|---------------------------|---------------------------------------|
| SFP                       | Up to 2.67 Gbps · SM/MM · up to 80 km |
| SFP+                      | Up to 11.3 Gbps · SM/MM · up to 40 km |
| SFP28                     | 25.78 Gbps · SM/MM · up to 10 km      |
| QSFP+ (40G)               | 41.2 Gbps (4×10G) · up to 10 km       |
| QSFP28 (100G)             | 103.1 Gbps (4×25G) · up to 2 km       |
| QSFP-DD (200G/400G)       | 200G / 400G · up to 2 km              |
| Copper SFP                | 10/100/1000BASE-T · Cat 5 · 100 m     |
| Operating Temperature     | 0°C to 70°C                           |
| Compliance                | Relevant MSA, IEEE 802.3, RoHS/REACH  |

## ACTIVE COMPONENTS

## SFP+ 10G BIDI

*Full range of MSA-compliant transceivers from 1.25G to 400G across SFP and QSFP form factors.*

PDR optical transceivers deliver hot-pluggable, MSA-compliant 1.25G–400G connectivity with SFP to QSFP-DD form factors, DDM, and SM/MM, CWDM/DWDM/BiDi support.

### Key Features

- MSA-compliant form factors: SFP, SFP+, SFP28, QSFP+, QSFP28, QSFP-DD and Copper SFP
- Single-mode and multimode options with reach up to 80 km
- Digital Diagnostic Monitoring (DDM/DOM) on optical variants
- Aggregate data rates from 1.25 Gbps to 400 Gbps
- Hot-pluggable with duplex LC (SFP) and MPO/MTP (QSFP) receptacles
- CWDM, DWDM and BiDi variants available on request

### Applications

- Telecom backbone, metro Ethernet and long-haul DWDM
- Enterprise and campus LAN uplinks
- Data-centre server-to-switch and spine aggregation

# Technical Specifications

| Parameter / Specification | Value / Detail                        |
|---------------------------|---------------------------------------|
| SFP                       | Up to 2.67 Gbps · SM/MM · up to 80 km |
| SFP+                      | Up to 11.3 Gbps · SM/MM · up to 40 km |
| SFP28                     | 25.78 Gbps · SM/MM · up to 10 km      |
| QSFP+ (40G)               | 41.2 Gbps (4×10G) · up to 10 km       |
| QSFP28 (100G)             | 103.1 Gbps (4×25G) · up to 2 km       |
| QSFP-DD (200G/400G)       | 200G / 400G · up to 2 km              |
| Copper SFP                | 10/100/1000BASE-T · Cat 5 · 100 m     |
| Operating Temperature     | 0°C to 70°C                           |
| Compliance                | Relevant MSA, IEEE 802.3, RoHS/REACH  |

## ACTIVE COMPONENTS

# SFP+ 10G Dual Fiber

*Standard dual-fiber 10G transmission*

PDR optical transceivers deliver hot-pluggable, MSA-compliant 1.25G–400G connectivity with SFP to QSFP-DD form factors, DDM, and SM/MM, CWDM/DWDM/BiDi support.

## Key Features

- MSA-compliant form factors: SFP, SFP+, SFP28, QSFP+, QSFP28, QSFP-DD and Copper SFP
- Single-mode and multimode options with reach up to 80 km
- Digital Diagnostic Monitoring (DDM/DOM) on optical variants
- Aggregate data rates from 1.25 Gbps to 400 Gbps
- Hot-pluggable with duplex LC (SFP) and MPO/MTP (QSFP) receptacles
- CWDM, DWDM and BiDi variants available on request

## Applications

- Telecom backbone, metro Ethernet and long-haul DWDM
- Enterprise and campus LAN uplinks
- Data-centre server-to-switch and spine aggregation

## Technical Specifications

| Parameter / Specification | Value / Detail                        |
|---------------------------|---------------------------------------|
| SFP                       | Up to 2.67 Gbps · SM/MM · up to 80 km |
| SFP+                      | Up to 11.3 Gbps · SM/MM · up to 40 km |
| SFP28                     | 25.78 Gbps · SM/MM · up to 10 km      |
| QSFP+ (40G)               | 41.2 Gbps (4×10G) · up to 10 km       |
| QSFP28 (100G)             | 103.1 Gbps (4×25G) · up to 2 km       |
| QSFP-DD (200G/400G)       | 200G / 400G · up to 2 km              |
| Copper SFP                | 10/100/1000BASE-T · Cat 5 · 100 m     |
| Operating Temperature     | 0°C to 70°C                           |
| Compliance                | Relevant MSA, IEEE 802.3, RoHS/REACH  |

## ACTIVE COMPONENTS

# 1G SFP BIDI

*Cost-effective 1G bi-directional link*

PDR optical transceivers deliver hot-pluggable, MSA-compliant 1.25G–400G connectivity with SFP to QSFP-DD form factors, DDM, and SM/MM, CWDM/DWDM/BiDi support.

## Key Features

- MSA-compliant form factors: SFP, SFP+, SFP28, QSFP+, QSFP28, QSFP-DD and Copper SFP
- Single-mode and multimode options with reach up to 80 km
- Digital Diagnostic Monitoring (DDM/DOM) on optical variants
- Aggregate data rates from 1.25 Gbps to 400 Gbps
- Hot-pluggable with duplex LC (SFP) and MPO/MTP (QSFP) receptacles
- CWDM, DWDM and BiDi variants available on request

## Applications

- Telecom backbone, metro Ethernet and long-haul DWDM
- Enterprise and campus LAN uplinks
- Data-centre server-to-switch and spine aggregation

# Technical Specifications

| Parameter / Specification | Value / Detail                        |
|---------------------------|---------------------------------------|
| SFP                       | Up to 2.67 Gbps · SM/MM · up to 80 km |
| SFP+                      | Up to 11.3 Gbps · SM/MM · up to 40 km |
| SFP28                     | 25.78 Gbps · SM/MM · up to 10 km      |
| QSFP+ (40G)               | 41.2 Gbps (4×10G) · up to 10 km       |
| QSFP28 (100G)             | 103.1 Gbps (4×25G) · up to 2 km       |
| QSFP-DD (200G/400G)       | 200G / 400G · up to 2 km              |
| Copper SFP                | 10/100/1000BASE-T · Cat 5 · 100 m     |
| Operating Temperature     | 0°C to 70°C                           |
| Compliance                | Relevant MSA, IEEE 802.3, RoHS/REACH  |

## ACTIVE COMPONENTS

# 25G & 25G BiDi

*Optimized for 5G fronthaul*

PDR optical transceivers deliver hot-pluggable, MSA-compliant 1.25G–400G connectivity with SFP to QSFP-DD form factors, DDM, and SM/MM, CWDM/DWDM/BiDi support.

## Key Features

- MSA-compliant form factors: SFP, SFP+, SFP28, QSFP+, QSFP28, QSFP-DD and Copper SFP
- Single-mode and multimode options with reach up to 80 km
- Digital Diagnostic Monitoring (DDM/DOM) on optical variants
- Aggregate data rates from 1.25 Gbps to 400 Gbps
- Hot-pluggable with duplex LC (SFP) and MPO/MTP (QSFP) receptacles
- CWDM, DWDM and BiDi variants available on request

## Applications

- Telecom backbone, metro Ethernet and long-haul DWDM
- Enterprise and campus LAN uplinks
- Data-centre server-to-switch and spine aggregation

# Technical Specifications

| Parameter / Specification | Value / Detail                        |
|---------------------------|---------------------------------------|
| SFP                       | Up to 2.67 Gbps · SM/MM · up to 80 km |
| SFP+                      | Up to 11.3 Gbps · SM/MM · up to 40 km |
| SFP28                     | 25.78 Gbps · SM/MM · up to 10 km      |
| QSFP+ (40G)               | 41.2 Gbps (4×10G) · up to 10 km       |
| QSFP28 (100G)             | 103.1 Gbps (4×25G) · up to 2 km       |
| QSFP-DD (200G/400G)       | 200G / 400G · up to 2 km              |
| Copper SFP                | 10/100/1000BASE-T · Cat 5 · 100 m     |
| Operating Temperature     | 0°C to 70°C                           |
| Compliance                | Relevant MSA, IEEE 802.3, RoHS/REACH  |

## ACTIVE COMPONENTS

# 400G

*Next-generation high-capacity throughput*

PDR optical transceivers deliver hot-pluggable, MSA-compliant 1.25G–400G connectivity with SFP to QSFP-DD form factors, DDM, and SM/MM, CWDM/DWDM/BiDi support.

## Key Features

- MSA-compliant form factors: SFP, SFP+, SFP28, QSFP+, QSFP28, QSFP-DD and Copper SFP
- Single-mode and multimode options with reach up to 80 km
- Digital Diagnostic Monitoring (DDM/DOM) on optical variants
- Aggregate data rates from 1.25 Gbps to 400 Gbps
- Hot-pluggable with duplex LC (SFP) and MPO/MTP (QSFP) receptacles
- CWDM, DWDM and BiDi variants available on request

## Applications

- Telecom backbone, metro Ethernet and long-haul DWDM
- Enterprise and campus LAN uplinks
- Data-centre server-to-switch and spine aggregation

# Technical Specifications

| Parameter / Specification | Value / Detail                        |
|---------------------------|---------------------------------------|
| SFP                       | Up to 2.67 Gbps · SM/MM · up to 80 km |
| SFP+                      | Up to 11.3 Gbps · SM/MM · up to 40 km |
| SFP28                     | 25.78 Gbps · SM/MM · up to 10 km      |
| QSFP+ (40G)               | 41.2 Gbps (4×10G) · up to 10 km       |
| QSFP28 (100G)             | 103.1 Gbps (4×25G) · up to 2 km       |
| QSFP-DD (200G/400G)       | 200G / 400G · up to 2 km              |
| Copper SFP                | 10/100/1000BASE-T · Cat 5 · 100 m     |
| Operating Temperature     | 0°C to 70°C                           |
| Compliance                | Relevant MSA, IEEE 802.3, RoHS/REACH  |

## ACTIVE COMPONENTS

# 40G

*Reliable mid-tier aggregation*

PDR optical transceivers deliver hot-pluggable, MSA-compliant 1.25G–400G connectivity with SFP to QSFP-DD form factors, DDM, and SM/MM, CWDM/DWDM/BiDi support.

## Key Features

- MSA-compliant form factors: SFP, SFP+, SFP28, QSFP+, QSFP28, QSFP-DD and Copper SFP
- Single-mode and multimode options with reach up to 80 km
- Digital Diagnostic Monitoring (DDM/DOM) on optical variants
- Aggregate data rates from 1.25 Gbps to 400 Gbps
- Hot-pluggable with duplex LC (SFP) and MPO/MTP (QSFP) receptacles
- CWDM, DWDM and BiDi variants available on request

## Applications

- Telecom backbone, metro Ethernet and long-haul DWDM
- Enterprise and campus LAN uplinks
- Data-centre server-to-switch and spine aggregation

## Technical Specifications

| Parameter / Specification | Value / Detail                        |
|---------------------------|---------------------------------------|
| SFP                       | Up to 2.67 Gbps · SM/MM · up to 80 km |
| SFP+                      | Up to 11.3 Gbps · SM/MM · up to 40 km |
| SFP28                     | 25.78 Gbps · SM/MM · up to 10 km      |
| QSFP+ (40G)               | 41.2 Gbps (4×10G) · up to 10 km       |
| QSFP28 (100G)             | 103.1 Gbps (4×25G) · up to 2 km       |
| QSFP-DD (200G/400G)       | 200G / 400G · up to 2 km              |
| Copper SFP                | 10/100/1000BASE-T · Cat 5 · 100 m     |
| Operating Temperature     | 0°C to 70°C                           |
| Compliance                | Relevant MSA, IEEE 802.3, RoHS/REACH  |

## ACTIVE COMPONENTS

# Copper SFP+

*RJ-45 interface for twisted pair copper*

PDR optical transceivers deliver hot-pluggable, MSA-compliant 1.25G–400G connectivity with SFP to QSFP-DD form factors, DDM, and SM/MM, CWDM/DWDM/BiDi support.

## Key Features

- MSA-compliant form factors: SFP, SFP+, SFP28, QSFP+, QSFP28, QSFP-DD and Copper SFP
- Single-mode and multimode options with reach up to 80 km
- Digital Diagnostic Monitoring (DDM/DOM) on optical variants
- Aggregate data rates from 1.25 Gbps to 400 Gbps
- Hot-pluggable with duplex LC (SFP) and MPO/MTP (QSFP) receptacles
- CWDM, DWDM and BiDi variants available on request

## Applications

- Telecom backbone, metro Ethernet and long-haul DWDM
- Enterprise and campus LAN uplinks
- Data-centre server-to-switch and spine aggregation

# Technical Specifications

| Parameter / Specification | Value / Detail                        |
|---------------------------|---------------------------------------|
| SFP                       | Up to 2.67 Gbps · SM/MM · up to 80 km |
| SFP+                      | Up to 11.3 Gbps · SM/MM · up to 40 km |
| SFP28                     | 25.78 Gbps · SM/MM · up to 10 km      |
| QSFP+ (40G)               | 41.2 Gbps (4×10G) · up to 10 km       |
| QSFP28 (100G)             | 103.1 Gbps (4×25G) · up to 2 km       |
| QSFP-DD (200G/400G)       | 200G / 400G · up to 2 km              |
| Copper SFP                | 10/100/1000BASE-T · Cat 5 · 100 m     |
| Operating Temperature     | 0°C to 70°C                           |
| Compliance                | Relevant MSA, IEEE 802.3, RoHS/REACH  |

ACTIVE COMPONENTS

# Smart SFP Transceiver

*Integrated OAM (Operations, Administration, and Maintenance)*

Integrated OAM (Operations, Administration, and Maintenance)

## Key Features

- Integrated OAM (Operations, Administration, and Maintenance)
- Real-time performance monitoring
- Micro-loopback capability

## Applications

- Service Provider Edge
- Mobile Backhaul
- SLA Monitoring

## Technical Specifications

| Parameter / Specification | Value / Detail         |
|---------------------------|------------------------|
| Interface                 | 1G / 10G SFP           |
| Protocol                  | IEEE 802.3ah / 802.1ag |
| Diagnostics               | DDM/DOM                |
| Latency                   | Ultra-Low              |

## PASSIVE COMPONENTS

# SMPTE Cable Assembly

*Ruggedized, dirt-protected fiber assemblies engineered for HDTV broadcast infrastructure.*

Ruggedized, dirt-protected SM/MM fiber assemblies with all-metal housings, heavy-duty retention, and silicon gaskets, engineered for HDTV broadcast and outdoor OB applications

## Key Features

- Rugged all-metal connector housing for durability
- Silicon gasket protection against dust and dirt
- Supports single-mode (9/125  $\mu\text{m}$ ) and multimode (50/125  $\mu\text{m}$ , 62.5/125  $\mu\text{m}$ , OM3) fiber
- Customizable lengths from 0.5 m to 999 m
- Heavy-duty cable retention for secure long-term performance
- Wide connector range: SC, LC, FC, ST, MU, MTRJ, SMA, E2000
- Multiple jacket types: simplex, duplex, mini-zipcord, armored, FTTH

## Applications

- HDTV broadcast infrastructure
- Broadcast studio installations
- Outdoor field deployment and Outside Broadcast (OB) vans System integration

## Technical Specifications

| Parameter / Specification | Value / Detail                                            |
|---------------------------|-----------------------------------------------------------|
| Fiber Type                | Single-mode (9/125 μm); Multimode (50/125, 62.5/125, OM3) |
| Connector Options         | SC, LC, FC, ST, MU, MTRJ, SMA, E2000                      |
| Insertion Loss            | ≤ 0.30 dB typical; ≤ 0.50 dB max                          |
| Return Loss               | ≥ 50 dB UPC; ≥ 60 dB APC                                  |
| Length Options            | 0.5 m to 999 m (custom available)                         |
| Operating Temperature     | −40 °C to +75 °C                                          |

## PASSIVE COMPONENTS

# VFL — Visual Fault Locator

*Visual fault locator for pinpointing fiber breaks, bends and faults.*

The BML209 is a compact pen-style 650 nm Visual Fault Locator (VFL) that detects fiber faults, breaks, bends, and connector defects, with 1–30 mW output options for 3 km to 10+ km testing.

## Key Features

- 650 nm  $\pm 20$  nm visible red laser for instant fault and break localisation
- Dynamic detection distance: 3 to 5 km (1 mW) up to >10 km (20/30 mW variants)
- 2.5 mm universal adapter connector; optional 1.25 mm adapter available
- Four output power variants: 1 mW, 10 mW, 20 mW, and 30 mW for different reach requirements
- Dual operation modes: Continuous Wave (CW) and Pulsed (2 Hz) for identification
- Detects faults in connectors, splices, bends, breaks, and fiber splitter points

## Applications

- Fault localisation in FTTH/FTTB drop cables, indoor wiring, and patch cords
- Splice point inspection and macro-bend fault detection in outside-plant cables
- Verification of fiber continuity and connector end-face quality
- Fiber splitter, coupler, and passive component testing

# Technical Specifications

| Parameter / Specification | Value / Detail                           |
|---------------------------|------------------------------------------|
| Output Power              | >1 mW / >10 mW / >20 mW / >30 mW         |
| Dynamic Distance          | 3 to 5 km / 8 to 10 km / >10 km / >10 km |
| Wavelength                | 650 nm ±20 nm                            |
| Operation Modes           | Continuous Wave (CW) and Pulsed (2 Hz)   |
| Connector                 | 2.5 mm Universal adapter                 |
| Dimensions                | 180 mm (L) × 30 mm (diameter)            |
| Weight                    | 46 g                                     |

## CABLE MANAGEMENT DEVICES

# Optical Fiber Wall Mount Enclosure

*Wall-mount optical enclosure for protected indoor fiber termination.*

PDR Optical Fiber Wall Mount Enclosure is a lockable double-door fiber termination unit for Singlemode/Multimode networks, supporting ST, FC, SC & LC adapters, loaded or unloaded options, and four cable entry points for flexible installation.

## Key Features

- Scalable port capacity, available in 12, 24, 48, and 96 port configurations
- Four-point cable entry (two top, two bottom) for flexible fiber routing
- Supplied loaded (with adapters) or unloaded, installer configures as required
- Double-door construction with key lock for secure, access-controlled fiber termination
- Accepts Singlemode and Multimode ST, FC, SC, and LC adapter plates
- Compact wall-mount footprint eliminates the need for a full 19" cabinet

## Applications

- Fiber-to-the-Home (FTTH) and Fiber-to-the-Building (FTTB) premises termination
- Enterprise LAN and structured cabling fiber termination in server rooms
- Telecommunications network distribution points and cross-connect locations
- CATV and broadband fiber distribution in residential and commercial buildings

## Technical Specifications

| Parameter / Specification | Value / Detail                         |
|---------------------------|----------------------------------------|
| Port Capacity             | 12 / 24 / 48 / 96 ports                |
| Construction              | Double-door, lockable steel enclosure  |
| Adapter Compatibility     | ST, FC, SC, LC, Singlemode & Multimode |
| Cable Entry Points        | 4 (2 top, 2 bottom)                    |
| Configuration             | Loaded / Unloaded                      |
| Dimensions, 12 Port       | 300 mm (H) × 250 mm (W) × 70 mm (D)    |
| Dimensions, 24 Port       | 312 mm (H) × 305 mm (W) × 85 mm (D)    |
| Dimensions, 48 Port       | 454 mm (H) × 370 mm (W) × 87 mm (D)    |
| Dimensions, 96 Port       | 454 mm (H) × 370 mm (W) × 177 mm (D)   |
| Mounting                  | Wall mount                             |
| Access Control            | Key lock provision                     |

## ACTIVE COMPONENTS

# Bypass Switch

*Carrier-grade 1U optical bypass switch that reroutes traffic around a failed inline device in under 50 ms.*

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Carrier-grade 1U optical path protection that reroutes traffic around a failed inline device in under 50 ms via power, optical-power, and TCP/IP heartbeat monitoring, with up to 4 channels and dual AC/DC power.

## Key Features

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- Automatic bypass protection via power supply, optical power, and TCP/IP heartbeat monitoring — individually or in combination
- Up to 4 independent bypass channels in a single 1U 19-inch rack chassis
- LC/PC single-mode (9/125  $\mu\text{m}$ ) interfaces with  $\leq 1.5$  dB insertion loss and  $\geq 50$  dB return loss
- Dual power inputs: AC 85–264 V (50/60 Hz) and DC 36–72 V for full redundancy
- Sub-50 ms carrier-grade switching speed for transparent network failover
- Auto-Switch and Manual-Switch modes with configurable auto-return after an idle period
- Switching-log records reason, time, and result of every switching event

## Applications

- Inline node protection for network security appliances — firewalls, IPS, DPI, and flow-control devices
- FTTH/FTTB access network deployments
- Military, industrial, and mission-critical communication systems
- Repeater and optical amplifier protection in long-haul and metro networks
- Data centre interconnects and spine-leaf fabric resilience

## Technical Specifications

| Parameter / Specification | Value / Detail                         |
|---------------------------|----------------------------------------|
| Working Wavelength        | 1310 nm / 1550 nm (selectable)         |
| Fiber Type                | SM 9/125 μm (ITU-T G.652)              |
| Channels                  | Up to 4 independent channels           |
| Insertion Loss            | ≤ 1.5 dB                               |
| Return Loss               | ≥ 50 dB                                |
| Switching Time            | < 50 ms                                |
| Power Supply              | AC 85–264 V   DC 36–72 V (dual inputs) |
| Dimensions                | 19-inch rack, 1U height                |

## CABLE MANAGEMENT DEVICES

# Home Termination Box (HTB)

*Compact home termination box for neat FTTH subscriber-side fiber management.*

PDR Home Termination Box (HTB) is a compact wall-mounted enclosure for fiber termination, splicing, adapters, FICs, pigtails and cable storage, with bend-radius protection and splice/slack management for reliable ONT/CPE connections.

## Key Features

- Compact, low-profile design occupies minimal wall space during installation
- Simple, tool-free installation with included mounting screws and cable ties
- Accepts both standard SC and LC optical adapters
- Inner slack storage area maintains minimum bend radius compliance at all times
- Inner tray accommodates splice sleeves for clean in-box splicing
- Available in Loaded (with adapters) and Unloaded (adapter-free) configurations

## Applications

- Fiber-to-the-Home (FTTH) and Fiber-to-the-Building (FTTB) last-mile deployments
- Local Area Network (LAN) structured cabling termination points
- Optical CATV and broadband fiber distribution networks
- ONT patch cord interface in residential and small commercial premises

Technical Specifications

| Parameter / Specification | Value / Detail                |
|---------------------------|-------------------------------|
| Dimensions (L×W×H)        | 86 × 86 × 25 mm               |
| Fiber Count               | 2 Fiber                       |
| Colour                    | White                         |
| Material                  | ABS, PP                       |
| Insertion Loss            | < 0.3 dB                      |
| Return Loss               | ≥ 65 dB (APC) / ≥ 55 dB (UPC) |
| Adapter Compatibility     | SC, LC (standard form factor) |
| Configuration             | Loaded / Unloaded             |
| Standard                  | IEC 61754-4                   |

## ACTIVE COMPONENTS

# 1G SFP Dual Fiber

*Legacy compatible 1G module*

PDR optical transceivers deliver hot-pluggable, MSA-compliant 1.25G–400G connectivity with SFP to QSFP-DD form factors, DDM, and SM/MM, CWDM/DWDM/BiDi support.

## Key Features

- MSA-compliant form factors: SFP, SFP+, SFP28, QSFP+, QSFP28, QSFP-DD and Copper SFP
- Single-mode and multimode options with reach up to 80 km
- Digital Diagnostic Monitoring (DDM/DOM) on optical variants
- Aggregate data rates from 1.25 Gbps to 400 Gbps
- Hot-pluggable with duplex LC (SFP) and MPO/MTP (QSFP) receptacles
- CWDM, DWDM and BiDi variants available on request

## Applications

- Telecom backbone, metro Ethernet and long-haul DWDM
- Enterprise and campus LAN uplinks
- Data-centre server-to-switch and spine aggregation

# Technical Specifications

| Parameter / Specification | Value / Detail                        |
|---------------------------|---------------------------------------|
| SFP                       | Up to 2.67 Gbps · SM/MM · up to 80 km |
| SFP+                      | Up to 11.3 Gbps · SM/MM · up to 40 km |
| SFP28                     | 25.78 Gbps · SM/MM · up to 10 km      |
| QSFP+ (40G)               | 41.2 Gbps (4×10G) · up to 10 km       |
| QSFP28 (100G)             | 103.1 Gbps (4×25G) · up to 2 km       |
| QSFP-DD (200G/400G)       | 200G / 400G · up to 2 km              |
| Copper SFP                | 10/100/1000BASE-T · Cat 5 · 100 m     |
| Operating Temperature     | 0°C to 70°C                           |
| Compliance                | Relevant MSA, IEEE 802.3, RoHS/REACH  |

## SPECIALTY DRONES

# UAV Fiber Optic Spool

*Ultra-light UAV payout spool that dispenses bend-insensitive fiber cleanly at high speed, from 500 m to 50 km.*

Ultra-light optical fiber payout spool for tethered UAVs and unmanned vehicles, wound with bend-insensitive G.657.A2 single-mode fiber supplied in 17 standard lengths (500 m–50 km) with an FC interface.

## Key Features

- Ultra-light payout spool design optimized for tethered UAV and unmanned-vehicle fiber links
- Optional Kevlar coating for high strength and wear resistance in harsh environments
- Rugged black housing built for demanding field and airborne conditions
- Bend-insensitive G657.A2 single-mode fiber for reliable high-speed payout with minimal bend loss
- Rated 60 N pulling force for controlled, snag-free fiber dispensing under tension
- Made in India

## Applications

- Tethered UAV and drone fiber-optic communication and control links
- Unmanned surface and underwater vehicles (USV/ROV) and unmanned vessels
- Rapid field-deployable and emergency fiber-optic communication links
- Defence, ISR and loitering / FPV unmanned systems requiring jam-resistant fiber tethers
- Underwater exploration, tunnels, pipelines and underground culvert deployment
- Industrial and research applications demanding lightweight, high-strength fiber payout

# Technical Specifications

| Parameter / Specification | Value / Detail                                                   |
|---------------------------|------------------------------------------------------------------|
| Weight                    | 80 g/km                                                          |
| Pulling Force             | 60N                                                              |
| Available Lengths         | 500 m to 50 km (17 standard variants — see Ordering Information) |

## SPECIALTY DRONES

# Sky Unit

*Airborne fiber optical terminal that sends camera video and flight-controller telemetry down a single fiber at 1.25 Gbps.*

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Aircraft-mounted transmit terminal that accepts camera composite video plus flight-controller telemetry and carries them over a single fiber at 1.25 Gbps, offered in four compact mounting sizes with a GH1.25 interface for direct integration into the drone's power and control harness.

## Key Features

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- Airborne (Type A) transmit terminal for tethered / FPV drone fiber links
- Accepts composite video input and flight-controller TTL telemetry
- GH1.25 (8P) back-panel connector carrying video, TTL serial and power
- Wide 5–12 V DC input for direct drone-power integration
- Single-fiber optical transmission at 1.25 Gbps over an FC interface
- Four mounting sizes (88×88 to 141×120 mm) for flexible airframe integration
- PWR and SYNC status LEDs for power and optical-sync diagnostics

## Applications

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- Airborne video/telemetry transmission for FPV and racing drones
- Long-range ISR and defence UAV downlinks
- Tethered-drone and loitering unmanned systems needing jam-resistant links
- Airframe integration paired with a fiber payout spool

# Technical Specifications

| Parameter / Specification | Value / Detail                |
|---------------------------|-------------------------------|
| Unit Type                 | Type A — Airborne Sky Unit    |
| Optical Transmission Rate | 1.25 Gbps                     |
| Optical Connector         | FC (single fiber)             |
| Serial / Telemetry        | TTL serial (Tx/Rx) via GH1.25 |
| Connector                 | GH1.25 (8P) back panel        |
| Power Budget              | 5–12 V DC (via GH1.25)        |

## PASSIVE COMPONENTS

# Fiber Optic Adapter

*Precision bulkhead adapters that join like connectors for low-loss optical continuity.*

Precision passive coupler that joins two like fiber optic connectors end-to-end across a bulkhead, patch panel or enclosure, using a zirconia-ceramic or phosphor-bronze split sleeve available for all common connector standards in SM and MM.

## Key Features

- Couples two like connectors (SC, LC, FC, ST, MU, E2000, MT-RJ) for optical continuity across a panel
- Low insertion loss  $\leq 0.2$  dB for minimal signal degradation
- Available in simplex, duplex, and quad configurations
- Compatible with both single-mode (OS1/OS2) and multimode (OM1-OM4) fiber
- High-precision zirconia-ceramic or phosphor-bronze split alignment sleeve for accurate ferrule alignment
- High return loss  $\geq 50$  dB (UPC) and  $\geq 60$  dB (APC) for single-mode
- UPC and APC polish options to match connector end-faces
- Plug-and-play installation with no special tools;  $\geq 1000$  mating cycles over  $-40$  °C to  $+75$  °C

## Applications

- Fiber-to-the-Home (FTTH) and Fiber-to-the-Building (FTTB) network deployments
- Data centre cross-connects and enterprise LAN cabling
- Telecommunications patch panels, ODFs, and bulkhead interconnects
- Cable TV (CATV) and broadband infrastructure

## Technical Specifications

| Parameter / Specification | Value / Detail                              |
|---------------------------|---------------------------------------------|
| Connector Types           | SC, LC, FC, ST, MU, E2000, MT-RJ            |
| Configuration             | Simplex / Duplex / Quad                     |
| Fiber Type                | Single-mode (OS1/OS2) / Multimode (OM1-OM4) |
| Polish                    | UPC / APC                                   |
| Durability                | ≥ 1000 mating cycles                        |

## TEST AND MEASUREMENT EQUIPMENT &gt; VIDEOSCOPE

# WiFi Wireless Fiber Endface Microscope

*Wireless endface microscope with automatic pass/fail analysis on Windows, Android, and iOS.*

Wireless/USB-C fiber endface inspector with automatic pass/fail analysis

## Key Features

- WiFi and USB Type-C data transfer with real-time video and automatic pass/fail analysis
- 0.42  $\mu\text{m}$  resolution, 5 MP CMOS sensor, 10X optical magnification, <1  $\mu\text{m}$  particle detection
- Built-in rechargeable Li-ion battery, up to 5 hours; usable while charging
- Compatible with Windows, Android, and iOS
- More than 50 adapter types including MPO/MTP; custom adapters available
- White LED lighting for low-light and dark environments

## Applications

- 5G optical network construction and maintenance
- High-reliability fiber connections
- Endface inspection before OTDR testing
- Data centre connector inspection
- Laboratory and manufacturing test

## Technical Specifications

| Parameter / Specification | Value / Detail                |
|---------------------------|-------------------------------|
| Resolution                | 0.42 µm                       |
| Image Sensor              | 5-megapixel CMOS              |
| Optical Magnification     | 10X                           |
| Field of View             | 512 µm × 384 µm               |
| Signal Output             | WiFi / USB Type-C             |
| Battery Life              | 5 hours (rechargeable Li-ion) |

## TEST AND MEASUREMENT EQUIPMENT

# Optical Fiber Identifier PDR4406F

*Three-in-one fiber identifier with built-in 10 mW VFL and optical power meter for live-fiber work.*

Three-in-one field instrument combining non-intrusive macro-bend fiber ID (800–1700 nm), a 10 mW 650 nm VFL, and a –50 to +26 dBm optical power meter across six calibrated wavelengths.

## Key Features

- Three-in-one instrument: Fiber Identifier, 10 mW Visual Fault Locator, and Optical Power Meter
- Wide wavelength range 800–1700 nm; fits 250 µm / 900 µm / 2 mm / 3 mm fibers
- Low pass-through insertion loss: ≤ 1.0 dB (250/900 µm), ≤ 1.5 dB (2/3 mm)
- Compact 230 × 43 × 36 mm, ~200 g, runs on 2× AA alkaline batteries
- Non-destructive macro-bending identification — no service interruption on live fibers
- Detects 270 Hz, 1 kHz, and 2 kHz modulated tones for specific fiber identification
- OPM range –50 to +26 dBm, calibrated at 850/1300/1310/1490/1550/1625 nm

## Applications

- FTTH and FTTB network installation and maintenance
- CATV and broadband infrastructure testing and troubleshooting
- Field restoration and emergency fault diagnosis on live optical networks
- Telecom carrier fiber identification, power measurement, and fault tracing
- Data centre active fiber tracing and link budget verification

## Technical Specifications

| Parameter / Specification   | Value / Detail                               |
|-----------------------------|----------------------------------------------|
| Wavelength Range            | 800 – 1700 nm                                |
| Applicable Fiber Diameters  | 250 µm / 900 µm / 2.0 mm / 3.0 mm            |
| Pass-Through Insertion Loss | ≤ 1.0 dB (250/900 µm); ≤ 1.5 dB (2.0/3.0 mm) |
| Identified Signal Types     | 270 Hz / 1 kHz / 2 kHz (modulated)           |
| VFL                         | 650 nm, 10 mW, 2.5 mm universal              |
| OPM Measurement Range       | –50 to +26 dBm                               |
| Power Supply                | 2× AA (1.5 V alkaline)                       |

ACTIVE COMPONENTS > OPTICAL LINE PROTECTION SYSTEM (OLPS)

# test product

*Paens vito valde vestigium ulterius delinquo officia absens acidus.*

---

ACTIVE COMPONENTS > ACTIVE OPTICAL CABLE (AOC)

test2

test

ACTIVE COMPONENTS > ACTIVE OPTICAL CABLE (AOC)

test3

test3

ACTIVE COMPONENTS

NEWTTEST 123

TESTTT

des1. des2

Key Features

- key1
- key2

Applications

- app1
- app2

Technical Specifications

| Parameter / Specification | Value / Detail |
|---------------------------|----------------|
| Cable Type                | 123            |
| Power Budget              | abc            |

ACTIVE COMPONENTS

sdfgh

CABLE MANAGEMENT DEVICES

fg

# hjk

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ACTIVE COMPONENTS

dfg

---

PASSIVE COMPONENTS > ADAPTERS

# TESTNEW

TAGLINE

des

## Key Features

- key

## Applications

- app

## Technical Specifications

| Parameter / Specification | Value / Detail |
|---------------------------|----------------|
| Cable Type                | 123            |

CABLE MANAGEMENT DEVICES > FIBER TERMINATION BOX (INDOOR)

789

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PASSIVE COMPONENTS > CABLE ASSEMBLY

# NEW TEST 456

*TAGLINE*

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